

Review Article

The transformative capacity of public sector organisations in sustainability transitions

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ABSTRACT

Public sector organisations (PSOs) such as municipalities, regulatory bodies, and public utilities are key actors in sustainability transitions. However, the conceptualisation of their transformative capacity is underdeveloped, as several strands of literature pay attention to the topic but remain disjointed. The article takes stock of the literature, reviewing the strengths and weaknesses of the main approaches. Building from there, the article proposes a higher-order holistic conceptual framework based on three constitutive elements: roles, resources, and abilities. We conceptualise the transformative capacity of a public sector organisation as the interaction between its roles, resources, and abilities in the enactment of its transformative agency. This framework offers a granular analytical approach for future empirical studies, acknowledging the diversity of PSOs and sustainability transitions pathways. It also offers practical clues for more targeted efforts to build transformative capacity in PSOs, which is crucial for accelerating transitions.

1. Introduction

Green sustainability transitions^h are complex processes that require purposive and directional action to transform current socio-

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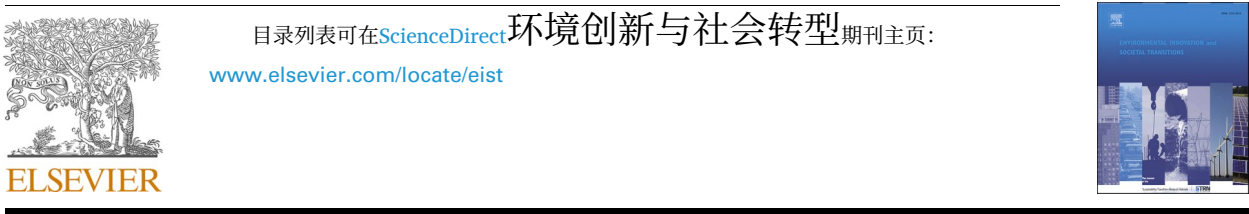
^h This article uses the term ‘sustainability transition’ to refer specifically to green dimension of sustainability transitions.

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综述文章

公共部门组织在可持续转型中的变革能力

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摘要

公共部门组织（如市政机构、监管机构和公用事业单位）是可持续转型的关键行动者。然而，对其转型能力的概念化尚不成熟，因为多支文献虽关注该主题但仍相互脱节。本文梳理文献，评述各主要方法的优缺点。在此基础上，本文提出一个基于三个构成要素（角色、资源和能力）的高阶整体概念框架。我们将公共部门组织的转型能力概念化为其角色、资源和能力在实施转型能动性过程中的互动。该框架为未来实证研究提供了精细化的分析方法，承认公共部门组织和可持续转型路径的多样性。它还为更有针对性地构建公共部门组织转型能力提供了实践线索，这对加速转型至关重要。

1. 引言

绿色可持续转型^h是复杂的过程，需要有目的、有方向的行动来转变当前的社会

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technical systems. These transitions are inherently linked to the capacity of various organisations to undertake transformative action as purposeful change agencies actively seeking to overcome the numerous barriers and lock-ins in the process of path-breaking transformation.

This article focuses specifically on the transformative capacity of public sector organisations (PSOs), with overall functions to serve the public interest and deliver public value (Moore 1995; Bozeman 2007). PSOs are a central building block of the state, implementing governmental policies in many different forms, scales, and levels of intervention. Hence, PSOs are a linchpin between the collective ambitions of green sustainability targets, and the reality of implementing them. To deliver, PSOs have to work towards overcoming the heavy inertia that characterises the current unsustainable modes of production and consumption. However, the likelihood of performing this pivotal function depends on each PSO’s transformative capacity. Despite the literature’s acknowledgment of the relevance of transformative capacity, the concept needs further conceptualisation.

Three strands in the recent literature have directed attention to this matter, while emphasising different important aspects.

The literature on public governance and public administration has increasingly paid attention to the problem-solving capacity of the modern state (Lodge and Wegrich 2014), focusing on policy capacity in a multi-level context (Wu et al., 2018). Particular attention has been paid to the complexity of grand challenges as wicked problems that need collective problem-solving (Termeer et al., 2016; Head 2018). However, this literature looks more into questions about policy consistency and the overall capacity of the state, rather than on the specific transformative capacity of PSOs in sustainability transitions.

The **innovation policy literature** has paid attention to public innovation in mission-oriented agencies, a subset of PSOs. It has also started to consider the dynamic capabilities of the state’s entrepreneurial initiatives (Breznitz, Ornston et al. 2018; Kattel and Mazzucato 2018; Kattel, Drechsler et al. 2022), and the context of transformative policy mixes (Rogge and Reichardt 2016; Weber and Rohrer 2012; Rogge, Pfluger et al. 2020). However, this literature has so far examined national mission agencies, without considering the capacity of other crucial PSOs for green innovation and systemic transformations, such as local public governments or publicly controlled utilities.

The **sustainability transition literature** has been increasingly focusing on change agency in the process of sustainability transitions (Köhler, Geels et al. 2019; Kivimaa, Boon et al. 2019; Geels 2020), particularly in urban contexts (Wolfram 2016; Castan Broto, Trencher et al. 2019; Hölscher and Frantzeskaki 2020; Peris-Blanes, Segura-Calero et al. 2022). This literature studies how transformative capacity is exercised in a distributed manner, which is key to the overall governance of sustainability transitions. With some exceptions, the literature tends to focus on the system level (mostly looking at governance or policy capacity overall) rather than specifically examining PSOs as key implementing actors exercising transformative action.

Taken together, these three strands represent a growing body of literature with highly relevant contributions. However, this emerging field remains fragmented and disjointed when it comes to defining the transformative capacity of PSOs in sustainability transitions. Hence, there is a need to take stock of these recent advances, searching for their strengths and weaknesses, and building further from them in order to advance a higher-order conceptual framework that moves forward this relevant research area (Haddad, Nakić et al. 2022; Duit 2023).

Consequently, this article has two main objectives. Firstly, we do an in-depth review of the main approaches in the three strands of literature. We analyse how scholars have dealt conceptually with the notion of *capacity*, distil their key insights, and highlight the strengths and shortcomings of each conceptual approach. Secondly, we move the research agenda forward by proposing a holistic conceptual framework for PSO transformative capacity in sustainability transitions. By bringing disjointed aspects together, elaborating on core aspects, and theorising further on the nature of interactions in sustainability transitions, we propose a higher-order conceptual framework that allows for a more granular understanding of transformative capacity in future empirical studies.

With these objectives in mind, this article addresses the following research questions: What are the various conceptual approaches and key aspects in studies about the capacity of PSOs in sustainability transitions? What are the key features of a higher-order conceptual framework about the transformative capacity of PSOs in sustainability transitions that can advance empirical studies on this crucial research agenda?

The article proceeds as follows. The next section explains the methodological approach for this literature review. Section 3 reviews, in-depth each of the three strands of literature mentioned above, identifying conceptual trends in core themes, as well as their respective strengths and weaknesses. Section 4 takes stock of the findings of the literature review and focuses on the key insights that emerge across the three strands. From these key insights, Section 5 conceptualises three constitutive elements of PSOs transformative capacity, namely, roles, resources, and abilities, into a higher-order framework. Section 6 suggests three relevant theoretical considerations: the interlinkages between roles, resources, and abilities; the co-evolution between PSOs’ transformative capacity and the processes/pathways of sustainability transitions; and the building of PSOs’ transformative capacity. Section 7 concludes by answering the research questions, considering the novel contributions of the higher-order conceptual framework proposed here, suggesting future lines for an empirical research agenda, and considering the practical implications for more targeted efforts to build the transformative capacity of PSOs given the urgent need to accelerate sustainability transitions.

2. Methodology and data

In this article, we conduct a scope review (Arksey and O’Malley 2005; Pham et al. 2014), which other authors call theoretical review (Paré, Trudel et al. 2015), narrative review (Petticrew and Roberts 2008), or meta narrative review (Gough, Thomas et al. 2012); a type of review that reviews the most important concepts and conceptual approaches.

This type of review “goes beyond merely assembling and describing past work. The primary contribution and value of this type of review lies in its ability to develop novel conceptualisations or extend current ones by identifying and highlighting knowledge gaps

技术系统。这些转型本质上与各组织作为有目的的变革机构，在突破路径的转型过程中积极寻求克服诸多障碍和锁定的能力相关联。

本文专门关注公共部门组织（PSOs）的转型能力，这些组织的总体职能是服务公共利益和实现公共价值（Moore 1995；Bozeman 2007）。PSO是国家的基本组成部分，以多种形式、规模和干预层面实施政府政策。因此，PSO是绿色可持续目标的集体雄心与实施现实之间的关键纽带。为实现目标，PSO必须努力克服当前不可持续的生产和消费模式所具有的严重惯性。然而，履行这一关键职能的可能性取决于每个PSO的转型能力。尽管文献已认识到转型能力的相关性，但该概念仍需进一步概念化。

近期文献中的三条研究线索 已关注此问题，同时强调了不同的重要方面

关于公共治理和公共行政的文献日益关注现代国家的问题解决能力（Lodge and Wegrich 2014）， 聚焦于多层级的政策能力（Wu et al., 2018）。特别关注了重大挑战作为棘手问题的复杂性，需要集体解决问题（Termeer et al., 2016；Head 2018）。然而，这类文献更多探讨政策一致性和国家整体能力问题，而非PSO在可持续转型中的具体转型能力。

创新政策文献关注了使命导向机构（公共部门组织的一个子集）中的公共创新。它也开始考虑国家创业举措的动态能力（Breznitz, Ornston et al. 2018; Kattel and Mazzucato 2018; Kattel, Drechsler et al. 2022）， 以及变革性政策组合的背景（Rogge and Reichardt 2016; Weber andRohrer 2012; Rogge, Pfluger et al. 2020）。然而，到目前为止，这些文献研究了国家使命机构，但没有考虑其他对绿色创新和系统性转型至关重要的公共部门组织的能力，例如地方政府或公有公用事业。

可持续转型文献越来越关注可持续转型过程中的变革能动性（Köhler, Geels et al. 2019; Kivimaa, Boon et al. 2019; Geels 2020），尤其是在城市背景下（Wolfram 2016; Castan Broto,Trencher et al. 2019; Hölscher and Frantzeskaki 2020; Peris-Blanes, Segura-Calero et al. 2022）。这些文献研究了变革能力如何以分布式方式行使，这对可持续转型的整体治理至关重要。除少数例外，这些文献倾向于关注系统层面（主要是从整体上看治理或政策能力），而不是专门考察公共部门组织作为行使变革性行动的关键执行者。

综上所述，这三个流派代表了一个日益增长且贡献高度相关的文献体系。然而，在定义公共部门组织在可持续转型中的变革能力时，这一新兴领域依然零散且支离破碎。因此，有必要梳理这些最新进展，审视其优势与不足，并在此基础上进一步构建一个能够推动这一重要研究领域向前发展的更高层次概念框架（Haddad,Nakić et al. 2022；Duit 2023）。

因此，本文有两个主要目标。首先，我们对三个文献流派的主要方法进行深入回顾。我们分析学者们如何在概念上处理能力这一概念，提炼其关键见解，并指出每种概念方法的优势与不足。其次，我们通过提出一个关于公共部门组织在可持续转型中变革能力的整体概念框架，推动研究议程向前发展。通过将零散的方面整合起来，阐述核心要点，并对可持续转型中互动的本质进行进一步理论化，我们提出了一个更高层次的概念框架，以便在未来的实证研究中对变革能力进行更细致的理解。

基于这些目标，本文探讨以下研究问题：在可持续转型中，关于公共部门组织能力的研究有哪些不同的概念方法和关键方面？一个关于公共部门组织在可持续转型中变革能力的更高层次概念框架有哪些关键特征，能够推动这一重要研究议程的实证研究？

本文的结构如下。下一节解释本文综述的方法论路径。第3节深入评述上述三个文献分支，识别核心主题的概念趋势以及各自的优缺点。第4节总结文献综述的发现，重点关注三个分支中出现的核心见解。基于这些核心见解，第5节将公共部门组织变革能力的三个构成要素——角色、资源和能力——概念化为一个更高层次的框架。第6节提出三个相关的理论考虑：角色、资源和能力之间的相互联系；公共部门组织’ 变革能力与可持续转型过程/路径的共同演化；以及公共部门组织’ 变革能力的构建。第7节通过回答研究问题、考虑本文提出的更高层次概念框架的新贡献、建议未来实证研究议程的路线，以及在加速可持续转型的迫切需求下，考虑更有针对性地构建公共部门组织变革能力的实践意义来结束本文。

2. 方法论与数据

在本文中，我们进行了一项范围审查（Arksey and O’Malley 2005; Pham 等人 2014），其他作者称之为理论审查（Paré, Trudel 等人 2015）、叙述性审查（Petticrew 和 Roberts 2008）或元叙述审查（Gough, Thomas 等人 2012）；这种审查类型回顾了最重要的概念和概念性方法。

这类综述 “不仅仅是整理和描述过去的工作。其主要贡献和价值在于能够识别并突出知识空白，从而发展新的概念或扩展现有的概念

between what we know and what we need to know” and “to provide a context for identifying, describing, and transforming into a higher order of theoretical structure and various concepts, constructs or relationships” (Paré, Trudel et al. 2015:188).

To ensure the methodological quality, we paid particular attention to the rigour, internal consistency, and relevance of the process and outcomes of this review (Paré, Trudel et al. 2015). We provide a detailed account of these key matters and criteria in Appendix 1. Appendix 2 provides the full list of the publications reviewed. Our review process was undertaken in four steps, as illustrated in Fig. 1.

In Step 1, we identified 1604 publications in two bibliographic databases. Based on our expertise and on manually selecting 9 core publications in the field (see Appendix 2), we defined the search terms and the script to be run in the bibliographical database search. Appendix 1 explains Step 1 in detail: the search terms, the decisions in the search, the databases used, the time span, and the search script.

In Step 2, we arrived at our final selection of publications. First, we carefully screened the titles and abstracts of the 1604 publications and made a selection according to explicit inclusion and exclusion criteria that relate to our research goals (see the criteria in Appendix 1). The 226 pre-selected publications underwent a final selection process where we scrutinised their full texts (following the same criteria), which resulted in 112 publications. Ultimately 121 publications were selected for review: 112 selected from databases and 9 core publications selected manually.

In Step 3, we reviewed the publications one by one in a contextual manner, that is, we conducted a review of the concepts in the papers in relation to the stream of literature that the publication belongs to. The research team worked interactively, in several rounds of meetings and discussions. This was done the following way. Each researcher was assigned several publications for review; sub-groups of researchers were organised around the literature streams. In the first sub-step, the research team identified the main themes emerging from each literature strand. The research team discussed in several iterative meetings which themes were most relevant according to the goals of this research. In the second sub-step, we conducted a deep dive, reviewing in detail the concepts and their interactions in all selected publications. This process allowed us to work individually, in sub-groups, and collectively as a whole team.

Step 4 is the conceptual development, the final stage of this research endeavour. We identified the key insights emerging across the literature (Section 4). From those insights, we develop a higher-order conceptual framework, conceptualising three constitutive elements of transformative capacity, namely, roles, resources, and abilities (Section 5). Lastly we engaged in iterative theorising efforts around some specific considerations (Weick 1989) (Section 6). Step 4 was done in a iterative manner both within the research team and with external researchers in various international seminars, providing several rounds of in-depth feedback and suggestions (Alvesson and Sandberg, 2011).

3. Literature review

Different literature strands have dealt with several themes and approaches to capacity. We carry out our literature review in two steps. Firstly, we review each strand in depth, studying the way they have approached these concepts, and assessing their strengths and shortcomings. Secondly, in Section 4, we engage in a cross-sectional review of the key insights that emerge from them.

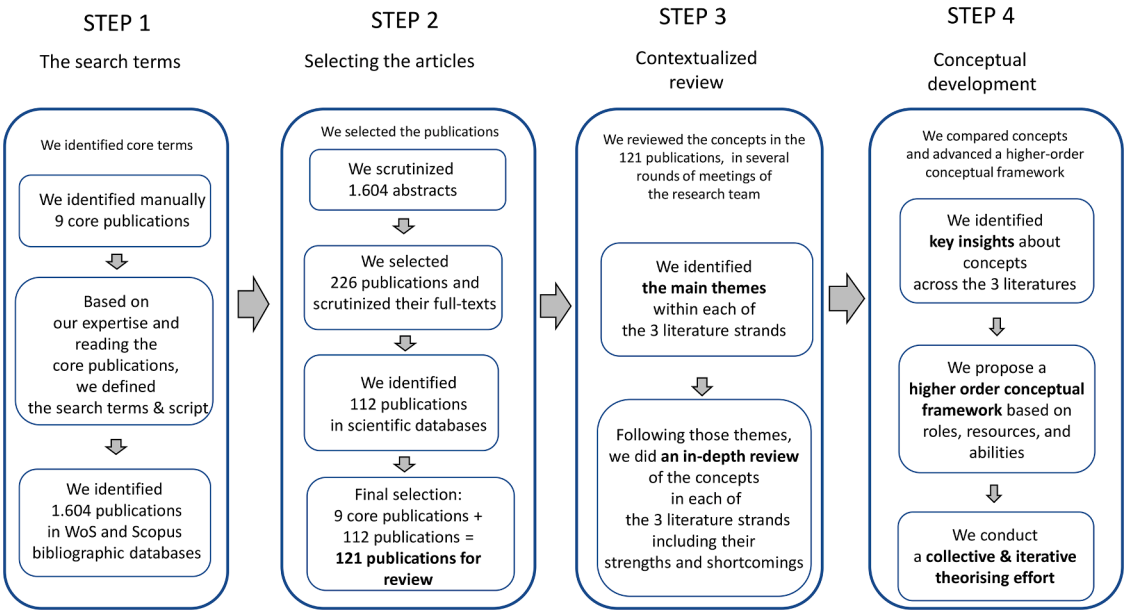


Fig. 1. The steps in the literature review process.

在已知与未知之间”以及“为识别、描述并转化为更高层次的理论结构以及各种概念、构念或关系提供背景” (Paré, Trudel et al. 2015:188)。

为确保方法论质量，我们特别关注了本综述过程和结果的严谨性、内部一致性和相关性 (Paré, Trudel et al. 2015)。我们在附录1中详细说明了这些关键事项和标准。附录2提供了所评阅文献的完整列表。我们的评阅过程分为四个步骤，如图1所示。

在第一步中，我们在两个文献数据库中识别出1604篇出版物。基于我们的专业知识以及手动挑选该领域的9篇核心出版物（见附录2），我们定义了搜索词和在文献数据库搜索中运行的脚本。附录1详细说明了第一步：搜索词、搜索决策、所用数据库、时间跨度以及搜索脚本。

在第二步中，我们得出了最终的出版物选择。首先，我们仔细筛选了1604篇出版物的标题和摘要，并根据与研究目标相关的明确纳入和排除标准（见附录1中的标准）进行了选择。226篇预选出版物经过最终筛选过程，我们对其全文进行了仔细审查（遵循相同标准），最终得到112篇出版物。最终共选择121篇出版物进行评阅：其中112篇来自数据库，9篇为核心出版物（手动选择）。

在步骤3中，我们以情境化方式逐一审阅了这些出版物，即根据每篇论文所属的文献流，对其中的概念进行了审查。研究团队通过多轮会议和讨论进行互动。具体做法如下：每位研究员被分配若干出版物进行审阅；子研究小组围绕文献流组织。在第一个子步骤中，研究团队识别了各文献脉络中出现的主要主题。研究团队通过多次迭代会议讨论哪些主题与本研究目标最相关。在第二个子步骤中，我们进行了深入探究，详细审阅了所有选定出版物中的概念及其相互作用。这一过程使我们能够以个人、子小组以及整个团队的形式协同工作。

步骤4是概念开发，也是本研究的最后阶段。我们识别了跨文献的关键见解（第4节）。基于这些见解，我们构建了一个高阶概念框架，概念化了变革能力的三个构成要素，即角色、资源和能力（第5节）。最后，我们围绕一些具体考虑进行了迭代理论化努力（Weick 1989）（第6节）。步骤4在研究团队内部以及与外部研究人员在多个国际研讨会中迭代进行，提供了多轮深入反馈和建议（Alvesson and Sandberg, 2011）。

3. 文献综述

不同的文献流派探讨了能力的多个主题和方法。我们分两步进行文献回顾。首先，我们深入研究每个流派，研究它们处理这些概念的方式，并评估其优势和不足。其次，在第4节中，我们对从中浮现的关键见解进行横向回顾。

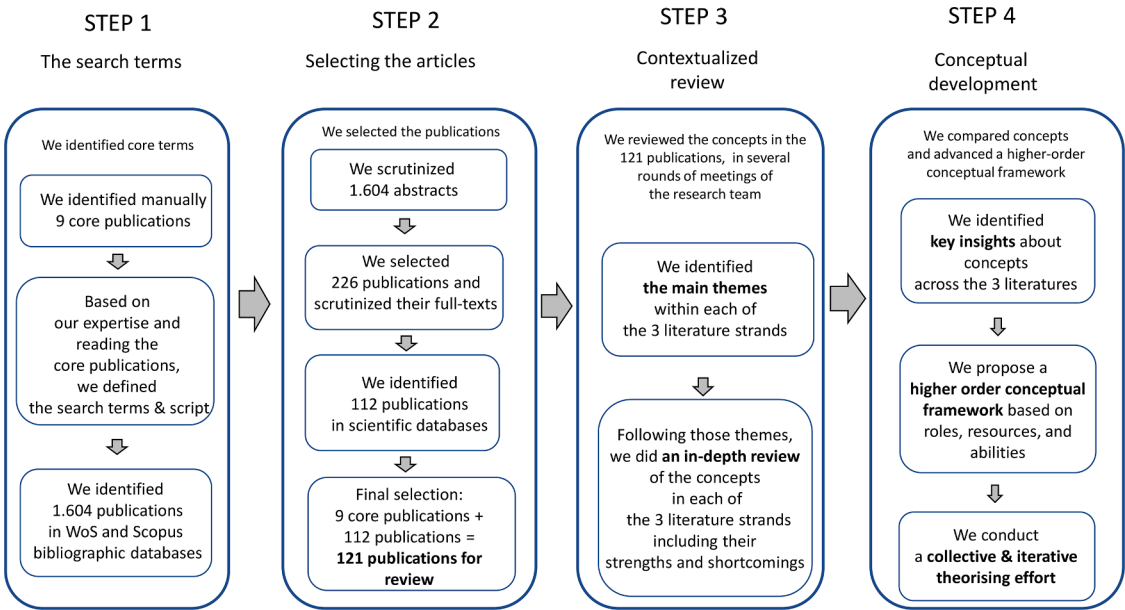


图1. 文献综述的步骤。

3.1. Public governance and public administration studies

The broad literature on public administration and governance deals with capacity in several interrelated themes, namely, the legacy of New Public Management and the debates around the role of the public sector and its capacity; the multi-level embeddedness of exercising capacity, including urban studies; the different dimensions of capacity as organisational abilities; and the transformative nature of public sector capacity. Our analysis scrutinizes these themes one-by-one in order to provide a structured approach.

Starting with the first theme, the New Public Management (NPM) approach emerged in the 1990s and was related to the idea of ‘reinventing government’ (Osborne and Gaebler 1992). This agenda aimed at transforming public administration by getting back to core tasks, by decoupling the roles of task-performing and task-steering, by increasing competition between the public and private sectors, by focusing on customers, and by increasing control over public administration performance. Capacity was mentioned sparsely, and only in terms of the public administration capacity to innovate, a strategy that in their perspective invariably involved outsourcing (Osborne and Plastrik 1997). Hence, the agenda of the NPM was about retrenching and transforming bureaucracy itself (Dunleavy and Hood 1994); it was not about public administration engaging in the transformation of socio-technical systems towards sustainability or other public goals. Therefore, the thrust of the NPM agenda sparked a debate about the roles, tasks, and overall purposes of the public sector that still resonates today (Moore 1995; Bozeman 2007; Seabrooke and Sending 2022). Whereas this latter Weberian tradition acknowledges the importance of agility, stability, and innovation in the public sector (Kattel, Drechsler et al. 2022), the extensive outsourcing of the past has been widely perceived as weakening the stance of public sector organisations (PSOs) by limiting severely their knowledge and other resources, a hurdle for them to engage in a transformative agenda towards solving complex collective problems (Mazzucato and Collington 2023).

The second theme in this literature stream is the multi-level embeddedness of exercising capacity. A pivotal publication in this theme defines ‘policy capacity’ as the ability of governments to marshal the necessary resources and to exercise them effectively to achieve desired policy objectives (Wu, Ramesh et al. 2018). Capacity is defined into three levels - systemic, organisational, and individual. The literature typically takes the starting point from the systemic-level, which refers to the overall capacity of policymaking and introduces considerations about organisational and individual levels. Policy capacity has been used in policy studies analyzing for example environmental policy, and in urban studies. In policy studies, capacity is considered in terms of the public administration ability to integrate environmental considerations into all policy sectors and levels (Ross and Dovers 2008), which requires the development of strategic, procedural, and evaluative abilities in the organisation of public administration. Likewise, urban studies look at capacity from a multi-level perspective, emphasizing that urban environmental performance is related to local governments’ capacity in terms of their different types of resources (their decentralized local mandates, their access to international networks of local governments, their staff, and finances, etc.) (Takao 2012; Takao 2017; Kim and Yoon 2018).

The third theme in this literature strand is about the different dimensions of capacity as organisational abilities. Following this approach some studies emphasize the relevance of analytical capacity of public administration in terms of its ability to analyse and make sense of complex data and information, which is key for environmental policy outcomes and evidence-based policymaking (Sugiyama and Muto 2022; Oliphant and Howlett 2010). Other studies emphasize the relevance of learning capacity as the organisational ability of public administration to learn from its own and from others’ experiences, constantly incorporating the new knowledge in policy making (Fiorino 2001). The ability of public administration to manage dilemmas, balance trade-offs, accommodate value conflicts, and deal with uncertainty among stakeholders in regulatory enforcement of green policies is another ability mentioned in the literature (Stenzel and Frenzel 2008; Kay and Ackrill 2012; Staricco and Buraschi 2022; van Popering-Verkerk, Molenveld et al. 2022). Managerial or operational capacity is another ability mentioned recurrently, referring to the internal human resources, as well as financial means, which are key in the delivery of every-day practices (Homsy and Warner 2015; Wu, Ramesh et al. 2018). Last but not least, another dimension mentioned in this literature is the political capacity (also named civic capacity) of the public administration, which has to do with the political legitimacy and societal trust as a key resource in any socio-political system, and the public administration’s ability to use this resource for engaging with society, mobilizing societal participation (Wu, Ramesh et al. 2018).

The fourth theme in this literature strand is the transformative nature of public administration capacity. Whereas the Weberian tradition emphasizes that the role of the public sector is to reduce uncertainties, a new Weberian approach acknowledges the complexity of social problems and new forms of public action (Kettl 2021), which depend on the capacity of public administration in terms of its context-dependent resources and roles (Peters 2015), and on the dynamic capabilities of bureaucracies as organisations (Kattel, 2022) (see Section 3.2). Following from that, a series of more recent studies focus on transformative and innovation capacity. Some authors point at reflexivity, inclusivity, adaptability, and creativity as key elements for transformative policy capacity. They define transformative policy capacity as the ability of policy actors to initiate and steer fundamental changes in the structures and dynamics of socio-ecological systems (Förster, Downsborough et al. 2021). This transformative capacity means that governments have more directional and strategic approaches, and that they are engaging in more collaboration with the business sector (Kuzemko 2022) and civil society organisations. Following from that remark, some authors in the new urban governance approach emphasize that transformative agency of PSOs has to do with the co-production of urban sustainable solutions, engaging and empowering the local communities and stakeholders (Novalia, Rogers et al. 2020). While other studies acknowledge that certain critical situations - like natural disasters related to climate change - might trigger the mobilization of transformative capacity towards path-breaking policy change (Nohrstedt 2022).

Taken together, the literature on public administration and governance acknowledges PSOs’ challenges to deliver solutions to complex problems. This literature has tended to identify specific dimensions such as managerial, fiscal, and civic capacity (Homsy and Warner 2015); analytical, operational and political (Wu et al. 2018); or analytical, delivery, coordination and regulatory capacity

3.1. 公共治理与公共行政研究

关于公共行政和治理的广泛文献涉及多个相互关联的主题，包括：新公共管理的遗产以及围绕公共部门角色及其能力的辩论；行使能力的多层次嵌入性，包括城市研究；作为组织能力的能力的不同维度；以及公共部门能力的变革性质。我们的分析逐一审视这些主题，以提供结构化的方法。

从第一个主题开始，新公共管理（NPM）方法兴起于20世纪90年代，与“重塑政府”的理念相关<style id=“3”>。（奥斯本和盖布勒，1992年）。这一议程旨在通过回归核心任务、将任务执行与任务指导的角色分离、增加公共与私营部门之间的竞争、关注客户以及加强对公共行政绩效的控制来改革公共行政。能力很少被提及，仅在公共行政创新能力方面有所涉及，这种策略在他们看来必然涉及外包（奥斯本和普拉斯特里克，1997年）。因此，NPM的议程是关于缩减和改革官僚机构本身（邓利维和胡德，1994年）；它并非关于公共行政参与社会技术系统向可持续发展或其他公共目标转型。因此，NPM议程的推动力引发了一场关于公共部门角色、任务和总体目标的辩论，这场辩论至今仍在继续（摩尔，1995年；博兹曼，2007年；西布鲁克和森丁，2022年）。虽然后一种韦伯式传统承认公共部门中敏捷性、稳定性和创新性的重要性（卡特尔、德雷克斯勒等人，2022年），但过去大范围的外包被广泛认为削弱了公共部门组织（PSOs）的地位，严重限制了其知识和其他资源，成为它们参与解决复杂集体问题的变革议程的障碍（马祖卡托和科林顿，2023年）。

该文献流中的第二个主题是行使能力的多层次嵌入性。该主题的一篇关键出版物将‘政策能力’定义为政府调动必要资源并有效运用以实现预期政策目标的能力（Wu, Ramesh等人，2018）。能力分为三个层次——系统级、组织级和个人级。文献通常以系统级为起点，系统级指的是政策制定的整体能力，并引入对组织和个人层面的考量。政策能力已被用于政策研究，例如环境政策分析以及城市研究中。在政策研究中，能力被视为公共行政将环境考量整合到所有政策部门和层级的能力（Ross and Dovers 2008），这要求公共行政组织发展战略、程序和评估能力。同样，城市研究从多层次视角审视能力，强调城市环境绩效与地方政府’的能力相关，这种能力体现在其不同类型的资源上（例如分散的地方授权、参与国际地方政府网络的机会、员工和财政等）（Takao 2012; Takao 2017; Kim and Yoon 2018）。

这一文献主题的第三个方面是关于不同类型的能力作为组织能力。按照这种思路，一些研究强调了公共行政的分析能力的重要性，即其分析并理解复杂数据和信息的能力，这对环境政策成果和循证决策至关重要（Sugiyama and Muto 2022; Oliphant and Howlett 2010）。其他研究则强调了学习能力的重要性，即公共行政从自身及他人的经验中学习，并不断将新知识纳入政策制定的组织能力（Fiorino 2001）。公共行政在绿色政策的监管执行中管理困境、平衡权衡、调和价值冲突以及应对利益相关者不确定性方面的能力，是文献中提到的另一种能力（Stenzel and Frenzel 2008; Kay and Ackrill 2012; Staricco and Buraschi 2022; van Popering-Verkerk, Molenveld et al. 2022）。管理或运营能力是另一个反复提及的能力，它涉及内部人力资源以及财政手段，这些在提供日常服务中至关重要（Homsy and Warner 2015; Wu, Ramesh et al. 2018）。最后但同样重要的是，这一文献中提到的另一个维度是公共行政的政治能力（也称为公民能力），这涉及政治合法性和社会信任，作为任何社会政治体系中的关键资源，以及公共行政利用这一资源与社会互动、动员社会参与的能力（Wu, Ramesh et al. 2018）。

该文献脉络中的第四个主题是公共行政能力的变革性质。韦伯传统强调公共部门的作用是减少不确定性，而新韦伯主义方法则承认社会问题的复杂性以及新的公共行动形式（Kettl 2021），这些行动依赖于公共行政在情境依赖资源和角色方面的能力（Peters 2015），以及官僚机构作为组织所具有的动态能力（Kattel, 2022）（见第3.2节）。由此，一系列近期研究聚焦于变革与创新能力。一些学者指出，反思性、包容性、适应性和创造力是变革性政策能力的关键要素。他们将变革性政策能力定义为政策行动者发起并引导社会生态系统结构和动态根本性变革的能力（Förster, Downsborough et al. 2021）。这种变革能力意味着政府采取更具方向性和战略性的方法，并与商业部门（Kuzemko 2022）及民间社会组织加强合作。基于此观点，新城市治理方法的一些学者强调，公共部门组织（PSO）的变革能动性 with 城市可持续解决方案的共同生产、地方社区和利益相关者的参与及赋权有关（Novalia, Rogers et al. 2020）。而其他研究则承认，某些危急情况——如与气候变化相关的自然灾害——可能触发变革能力的动员，推动路径突破性的政策变革（Nohrstedt 2022）。

总的来说，公共行政与治理领域的文献承认公共部门组织（PSO）在提供复杂问题解决方面面临的挑战。这些文献倾向于识别特定维度，如管理能力、财政能力和公民能力（Homsy and Warner 2015）；分析能力、运作能力和政治能力（Wu et al. 2018）；或分析能力、交付能力、协调能力和监管能力

(Lodge and Wegrich 2014). Often, this literature sees capacity largely as the set of internal and external resources to public organisations, which are mobilized by the PSO. This takes place in a multi-level governance context, where these PSOs are embedded, and their resources situated.

A possible limitation of these studies is their predominant focus on policy enforcement and outcomes within existing management and administrative paradigms. Policy consistency is most probably associated with the centrality of early discussions in the field about ‘environmental policy mainstreaming’ or ‘environmental policy integration’ (Jordan and Lenschow 2010) and about policy enforcement (Börzel and Buzogány 2019). With some exceptions, this strand of literature tends to understudy the importance of PSOs’ embeddedness in wider governance processes (beyond policy enforcement), as well as the organisational capacities related to transformative change. However, as we have seen in the review above, the literature is gradually addressing those shortcomings. Urban studies are bringing clearly to the fore the embeddedness of local governments’ administrations in the transitioning towards green urban systems; and the literature is increasingly acknowledging the active steering of the state towards transformation, and the relevance of the transformative capacity of PSOs.

3.2. Innovation studies

The field of innovation studies has also started to pay attention to the capacity of PSOs in the governance of green innovation and green transitions, mostly from the perspective of mission-oriented and/or transformative innovation policy (Haddad, Nakić et al. 2022), and/or from a broader perspective of the various roles of the sector action towards the transformation of socio-technical systems (Mazzucato 2015) (Borrás and Edler 2020). We have identified two specific themes in this literature strand. The first theme is the transformative capacity of PSOs seen as dealing with innovation processes, encompassing technological innovation as well as social innovation and public sector innovation; the second theme focuses on the importance of public administration self-organising its directional and purposeful action.

Weber and Rohrercher’s seminal article about transformative innovation redefines innovation policy beyond mere adaptation within existing paradigms and into a true transformative thrust, laying the ground for a new understanding of public sector capacity in transformation processes, particularly in green transitions (Weber and Rohrercher 2012). They identify systemic failures such as failures in directionality, demand articulation, policy coordination, and reflexivity, each elucidating different aspects of policy capacity and its implications for transformation. Particularly important are the requirements of policy coordination and reflexivity, which “are likely to raise the need to increase administrative capacity. This implies that the frequent claims for down-sizing the capacities of public administration, but also of the private sector, for managing the necessary policy coordination processes, in particular in the context of transformative change, need to be questioned” (Weber and Rohrercher 2012, 1044). Weber and Rohrercher emphasise the importance of public administration coordination and reflexivity capacity as processes and abilities that are needed to manage the complex and multi-level nature of these transformations. Hence, they consider the capacity of PSOs in terms of their abilities to enact transformation.

Along the same lines, Bos and Brown argue that the organisational capacity for transition policy programs depends on various factors, such as the organisational structure, culture, leadership, and resources. They also argue that organisational capacity can be enhanced by adopting a transition management approach, which involves envisioning, experimenting, learning, and scaling up innovative solutions (Bos and Brown 2014). Similarly, Isaksson and Hagbert examine the institutional capacity to integrate ‘radical’ perspectives on sustainability, which refer to the public sector adopting alternative and transformative visions and practices that challenge the dominant and unsustainable paradigms. They argue that the institutional capacity to integrate such ‘radical’ perspectives depends on existing norms, values, and rules, which can be enhanced by fostering a culture of innovation, collaboration, and experimentation in the public sector (Isaksson and Hagbert 2020).

The publications focusing on the second theme, on public administration’s self-organising and directional work in transformative innovation, tend to adopt a more strategic perspective. Most prominent in this approach is Kattel and Mazzucato’s use of the concept of *dynamic capabilities* in relation to public administration for implementing mission-oriented innovation policies addressing grand societal challenges (Kattel and Mazzucato 2018). They distinguish between the Schumpeterian tradition of dynamic capabilities in strategic management studies and the Weberian tradition of building public sector capacity through capable, autonomous bureaucracies. They propose merging these traditions, suggesting that developing dynamic capabilities as adaptive organising forms are essential for transformative action.

Taken together, the innovation studies literature strand tends to highlight the active role of the state in driving and shaping the transformative innovations seeking to address the systemic failures associated with grand challenges and green transitions (Anadon, Chan et al. 2016; Weber and Rohrercher 2012; Gieske, van Buuren et al. 2016). This perspective aligns well with the understanding of the dynamic nature of capacity in contemporary PSOs, and of state action for systemic change (Borrás and Edler 2020). Additionally, these studies refer to the concept of the dynamic capabilities of the PSOs, which focuses attention on the designed and acquired abilities of PSOs for engaging in transformative action (Mazzucato and Katel 2018; Breznitz et al. 2018). This literature outlines important approaches to how public sector entities can adapt and evolve to address emerging challenges and opportunities. However, this literature has tended to focus mostly on national innovation policy for executive agencies and other similar public organisations directly related to innovation policy, missing the wider picture of the multiplicity of PSOs currently engaged in green innovation for sustainability transitions, such as public utilities and local governments, and also underestimating the importance of the heterogeneous resources available to the PSOs.

(Lodge and Wegrich 2014)。通常，这类文献将能力主要视为公共组织内部和外部资源的总和，这些资源由PSO调动。这发生在多层次治理背景下，PSO嵌入其中，其资源也扎根于此。

这些研究的一个可能局限在于，它们主要关注现有管理和行政范式下的政策执行与结果。政策一致性很可能与该领域早期关于“环境政策主流化”或“环境政策一体化” (Jordan and Lenschow 2010)以及政策执行 (Börzel and Buzogány 2019) 的讨论中心地位有关。除少数例外，这类文献往往忽略公共部门组织在更广泛治理过程中的嵌入性（超越政策执行），以及与变革性变革相关的组织能力。然而，正如我们在上述综述中所见，相关研究正逐渐弥补这些不足。城市研究清晰地凸显了地方政府行政机构在向绿色城市系统转型中的嵌入性；同时，相关文献越来越多地承认国家对转型的积极引导，以及公共部门组织的变革能力的重要性。

3.2. 创新研究

创新研究领域也开始关注公共部门组织在绿色创新和绿色转型治理中的能力，主要从使命导向和/或变革性创新政策 (Haddad, Nakić et al.2022) 的视角，以及/或者从部门行动在推动社会技术系统转型中的多种角色 (Mazzucato 2015) (Borrás and Edler 2020) 的更广泛视角出发。我们在这条文献线索中识别出两个具体主题。第一个主题是公共部门组织作为处理创新进程（包括技术创新、社会创新和公共部门创新）的变革能力；第二个主题关注公共行政自我组织其方向性和目的性行动的重要性。

Weber和Rohrercher关于变革性创新的开创性文章将创新政策重新定义为超越现有范式内的简单适应，并进入真正的变革推动，为理解公共部门在转型过程中的能力（特别是绿色转型）奠定了基础 (Weber and Rohrercher 2012)。他们识别出系统性失败，如方向性、需求表达、政策协调和反思性等方面的失败，每一点都阐明了政策能力的不同方面及其对转型的影响。尤其重要的是政策协调和反思性的要求，这“可能会增加行政能力的必要性。这意味着，需要质疑那些频繁提出的缩减公共行政能力（以及私营部门能力）以管理必要政策协调过程的主张，特别是在变革性变化的背景下” (Weber and Rohrercher 2012, 1044)。Weber和Rohrercher强调，公共行政的协调和反思能力是管理这些复杂、多层面转型所需的过程和能力。因此，他们将公共部门组织的能力视为其实施转型的能力。

同理，Bos和Brown认为，转型政策方案的组织能力取决于多种因素，例如组织结构、文化、领导和资源。他们还指出，可以通过采用转型管理方法来增强组织能力，该方法涉及构思、实验、学习和推广创新解决方案 (Bos and Brown 2014)。类似地，Isaksson和Hagbert研究了整合可持续发展的“激进”视角的制度能力，这指的是公共部门采纳挑战主导及不可持续范式的替代性和变革性愿景与实践。他们认为，整合此类“激进”视角的能力取决于现有的规范、价值观和规则，而通过培养公共部门的创新、协作和实验文化可以增强这种能力 (Isaksson and Hagbert 2020)。

关注第二个主题（即公共行政在变革性创新中的自组织和方向性工作）的文献往往采用更具战略性的视角。其中最具代表性的是Kattel和Mazzucato将“动态能力”概念应用于公共行政，以实施应对重大社会挑战的任务导向型创新政策 (Kattel and Mazzucato 2018)。他们区分了战略管理研究中熊彼特传统的动态能力与通过能干、自主的官僚机构建设公共部门能力的韦伯传统。他们提出合并这些传统，认为发展作为适应型组织形式的动态能力对于变革行动至关重要。

总体而言，创新研究文献流派倾向于强调国家在推动和塑造旨在应对重大挑战和绿色转型相关系统性失灵的变革性创新中的积极作用 (Anadon,Chan et al. 2016; Weber and Rohrercher 2012; Gieske, van Buuren et al. 2016)。这一观点与当代公共部门组织中能力的动态性质以及国家行动促进系统性变革的理解相一致 (Borrás and Edler 2020)。此外，这些研究还提到了公共部门组织的动态能力概念，该概念关注公共部门组织为参与变革性行动而设计和获得的能力 (Mazzucato and Katel 2018; Breznitz et al. 2018)。这些文献概述了公共部门实体如何适应和发展以应对新出现的挑战和机遇的重要方法。然而，这些文献大多侧重于国家创新政策在直接与创新政策相关的执行机构和其他类似公共组织中的应用，忽略了当前参与绿色创新以促进可持续转型的众多公共部门组织的更广泛图景，如公共事业单位和地方政府，同时也低估了公共部门组织可获得的异质资源的重要性。

3.3. Sustainability transition studies

When dealing with the topic of the capacity of PSOs, the literature on sustainability transitions links several themes: the diversity of change actors, their interactions in power relations, and their different capabilities; the multi-level nature of transition processes; the contextualised capacities of actors across those levels; and the transformative capacity of entire urban systems.

The diversity of actors is a recurrent theme in sustainability transition studies. These actors include not only the niche innovators and the regime incumbents according to the MLP framework (Geels and Kemp 2012), but also the intermediaries, supporters, and opponents of transition. They have different roles, interests, values, and visions, and they operate at different scales, sectors, and stages of transition. Moreover, these actors are not static or homogeneous, but rather dynamic and heterogeneous, as they can shift their roles, identities, and alliances over time and across contexts (Hagbert and Malmqvist 2019). PSOs are generally perceived as one among those multiple actors, yet with some special relevance given the institutional nature of policy making (mainly at the regime level).

Very relevant for this literature is that the relative capacity of the multiple actors in a transition process is characterised by power interactions, opposite views, and resistance to change (Avelino and Wittmayer 2016). Power refers to the ability of actors to influence the behaviour or outcomes of other actors or systems, and it can be exercised through various means, such as coercion, persuasion, or participation. Power relations are not fixed or neutral, but rather contested and normative, as they reflect the distribution of resources, interests, and values among actors. Power relations might enable or constrain the capacity of actors to engage in a transition, and they can also be challenged or transformed by policymaking and implementation (Kuzemko et al. 2016). Moreover, power relations are vertical and horizontal, as they involve not only the interactions between different levels but also the interactions among actors within the same level, who may have opposite and/or different views and cognitive barriers/opportunities (Olazabal and Pascual 2015; Arts and Leroy 2006; Beck et al. 2021; Blythe et al. 2022; Boer 2013; Chaudhury, 2020; Hawkins et al. 2018).

This literature focuses on various dimensions of capacity. For example, *co-productive capacity* refers to the science-governance relations which might foster the mutual understanding, respect, and trust among different change actors in the system (van Kerkhoff and Lebel 2015), ultimately enhancing environmental governance capacity (van der Molen 2018). Other authors emphasise imagination and futuring; for example, Moore and Milkoreit (2020) discuss the role of imagination and creativity in building the capacity of actors and systems to envision and enact sustainable futures. In a similar vein, Pedde et al. (2019) suggest that scenarios can help build the collective capacity for societal learning, participation, and action towards sustainable transitions.

A specific sub-group of this literature looks into the *green dynamic capabilities* of organisations, particularly in governmental organisations (Arshad, Yu et al. 2023) and in publicly controlled utilities engaged in energy or water system transitions (Hartman, Gliedt et al. 2017; Lieberherr and Truffer 2015). Green dynamic capabilities are the specific abilities that enable PSOs to create and implement green innovations. Green dynamic capabilities are related to the culture, structure, resources, and learning processes inside these organisations, varying considerably across organisations. Studies about the innovation challenges of sustainability transitions suggest that utilities need to combine a capabilities and regime perspective, which means that they need to develop and mobilise their internal and external resources as well as their dynamic capabilities in order to change the dominant regime (van Welie, Truffer et al. 2019).

Some studies put emphasis on the national level, examining for example the overall national green innovation capacity (and its limits) (LaBelle 2017; Richter and Mundaca 2015; Meng and Wang 2022), or the capacity of developing countries for implementing green transitions (Konrad, van Deursen et al. 2021). At sub-national level, studies investigate the capacities and constraints in regional climate action and energy transitions (Royles and McEwen 2015; Pigott 2018; van Dijk, Wieczorek et al. 2022). These studies highlight the relevance of multi-level governance, devolution, and intergovernmental relations as enabling or limiting the capacity of sub-state governments to enact their change agency (Marquardt 2017).

Urban transformative capacity is a notion that has recently gained currency in the field of new urban governance studies, and it refers to the ability of urban systems to initiate, facilitate, and steer sustainability transitions in a deliberate and strategic manner (Wolfram et al., 2019). Transformative capacity is not a fixed attribute, but rather a systemic process that depends on the diversity and connectivity of urban actors, the availability and accessibility of resources and knowledge, the alignment and coherence of visions and goals, the legitimacy and accountability of governance processes, and the learning and innovation potential of urban experiments and initiatives (Hölscher, 2020; Spyridaki et al., 2020; Wolfram, 2018; Borgström 2019; Castan Broto et al. 2019; Peris-Blanes et al., 2022). Other studies about the capacity for sustainability transitions at the municipal level have revealed some gaps and challenges that need to be addressed. For instance, many studies note the lack of capacity of local governments to implement and enforce effective climate policies and regulations, especially when there are competing interests, institutional fragmentation, and resource constraints (Bryan, 2016; He and Hultman, 2021; Larson et al., 2017; Ryan, 2015; Ye et al., 2008).

Taken together, the sustainability transitions literature has a long tradition of looking into the transformation of socio-technical systems mainly at a systemic level, but also of focusing on change agency. It has studied power relations and tensions, acknowledging that some social groups and organisations might oppose change. Likewise, the recent approach to urban transformative capacity emphasises the relevance of urban systems. Thus, this literature is particularly strong in its process-oriented and systemic-context approach to transformative change. With some exceptions, it tends to focus on the system level and relatively less on the organisational level. Likewise, while this literature rightly points to the importance of the interaction between the public, private, and societal sectors, it nonetheless tends to underplay the relevance of the public sector in its ultimate function and goal of serving the public interest.

3.3. 可持续转型研究

在处理公共部门组织能力这一主题时，可持续转型文献将几个主题联系起来：变革行动者的多样性、他们在权力关系中的互动及其不同能力；转型过程的多层次性；这些层次中行动者的情境化能力；以及整个城市系统的变革能力。

行动者的多样性是可持续转型研究中一个反复出现的主题。这些行动者不仅包括根据MLP框架 (Geels and Kemp 2012) 的利基创新者和制度在位者，还包括转型的中介者、支持者和反对者。他们具有不同的角色、利益、价值观和愿景，并在转型的不同规模、部门和阶段中运作。此外，这些行动者并非静态或同质，而是动态和异质的，因为他们可以随着时间和背景的变化而转变自己的角色、身份和联盟 (Hagbert and Malmqvist 2019)。公共部门组织通常被视为这些多元行动者中的一员，但由于政策制定的制度性质（主要在制度层面），它们具有特殊的相关性。

与这一文献非常相关的是，转型过程中多元行动者的相对能力以权力互动、对立观点和变革阻力为特征 (Avelino and Wittmayer 2016)。权力指行动者影响其他行动者或系统行为或结果的能力，可以通过强制、说服或参与等多种方式行使。权力关系并非固定或中立，而是有争议且具有规范性的，因为它们反映了行动者之间资源、利益和价值观的分配。权力关系可能促进或限制行动者参与转型的能力，也可能通过政策制定和实施而受到挑战或转变 (Kuzemko et al. 2016)。此外，权力关系是纵向和横向的，因为它们不仅涉及不同层级之间的互动，还涉及同一层级内部行动者之间的互动，这些行动者可能持有对立和/或不同的观点以及认知障碍/机会 (Olazabal and Pascual 2015; Arts and Leroy 2006; Beck et al. 2021; Blythe et al. 2022; Boer 2013; Chaudhury, 2020; Hawkins et al. 2018)。

本文献关注能力的多个维度。例如，共同生产能力指的是科学-治理关系，这种关系可能促进系统中不同变革行动者之间的相互理解、尊重和信任 (van Kerkhoff and Lebel 2015)，最终提升环境治理能力 (van der Molen 2018)。其他作者强调想象力和未来构想；例如，Moore和Milkoreit (2020) 讨论了想象力和创造力在构建行动者和系统能力以构想和实现可持续未来方面的作用。类似地，Pedde等人 (2019) 提出，情景可以帮助构建集体能力，促进社会学习、参与和朝向可持续转型的行动。

本文献的一个特定分支探讨组织的绿色动态能力，特别是在政府组织 (Arshad, Yu et al. 2023) 以及参与能源或水系统转型的公用事业公司 (Hartman, Gliedt et al. 2017; Lieberherr and Truffer 2015) 中。绿色动态能力是使公共部门组织能够创造和实施绿色创新的特定能力。绿色动态能力与这些组织内部的文化、结构、资源和学习过程相关，在不同组织间差异显著。关于可持续转型创新挑战的研究表明，公用事业公司需要将能力视角与体制视角相结合，这意味着它们需要开发并动员其内部和外部资源以及动态能力，以改变主导体制 (van Welie, Truffer et al. 2019)。

一些研究强调国家层面，例如考察整体国家绿色创新能力（及其局限性） (LaBelle 2017; Richter and Mundaca 2015; Meng and Wang 2022)，或发展中国家实施绿色转型的能力 (Konrad, van Deursen et al. 2021)。在次国家级层面，研究考察了区域气候行动和能源转型中的能力与制约因素 (Royles and McEwen 2015; Pigott 2018; van Dijk, Wieczorek et al. 2022)。这些研究强调了多层次治理、权力下放和政府间关系作为赋能或限制次国家级政府发挥其变革能动性的相关性 (Marquardt 2017)。

城市转型能力是新城市治理研究领域近期流行的概念，指城市系统以有意识、战略性的方式启动、促进和引导可持续转型的能力 (Wolfram et al., 2019)。转型能力并非固定属性，而是一个系统性过程，取决于城市行动者的多样性和连通性、资源和知识的可用性与可及性、愿景和目标的一致性与连贯性、治理过程的合法性与问责制，以及城市实验和倡议的学习与创新潜力 (Hölscher, 2020; Spyridaki et al., 2020; Wolfram, 2018; Borgström 2019; Castan Broto et al. 2019; Peris-Blanes et al., 2022)。其他关于市级可持续转型能力的研究揭示了一些需要解决的差距和挑战。例如，许多研究指出地方政府缺乏实施和执行有效气候政策与法规的能力，尤其是在存在竞争利益、制度碎片化和资源限制的情况下 (Bryan, 2016; He and Hultman, 2021; Larson et al., 2017; Ryan, 2015; Ye et al., 2008)。

综合来看，可持续转型文献在系统层面研究社会技术系统转型方面历史悠久，同时也关注变革主体。它研究了权力关系和紧张局势，承认某些社会群体和组织可能反对变革。同样，最近关于城市变革能力的方法强调了城市系统的相关性。因此，这类文献在变革性变革的过程导向和系统背景方法方面尤为强大。除少数例外，它倾向于关注系统层面，相对较少关注组织层面。同样，尽管这类文献正确指出了公共、私营和社会部门之间互动的重要性，但它往往低估了公共部门在服务公共利益这一最终功能和目标中的相关性。

4. The findings: key insights across the literature

The literature review above brings forward two key insights, one about the different approaches to capacity and the second about the ‘transformativeness’ in transformative capacity.

4.1. Key insights about capacity

We observe that some of the reviewed publications use the term *capacity*, whereas others use *capability*; some refer to capacity in general terms (like *institutional capacity*) whereas others refer to specific dimensions of capacity (like *analytical capacity*). In this subsection, we examine how these terms have been conceptualised, identifying their overall background and, most importantly, how they relate to each other.

Capacity is the most extensively used term in the literature and we can identify at least two approaches: Capacity as a resource-based potential and capacity as action-oriented ability.

Capacity as resource-based potential

In the first approach, authors refer to capacity in terms of potential or manoeuvring space, as opportunities to be materialised (Olazabal and Pascual 2015), often in relation to resources (George and Reed 2016). Capacity is seen as the overall potential of an organisation as its license to operate given its resources, which are typically defined by its institutional positioning/situation in a specific context (Pelling, High et al. 2008). Hence, in that sense, the capacity of a public sector organisation refers to its overall potential according to its situated resources—such as political mandate, available financial resources, legitimacy, or human resources—allowing for autonomous action (Royles and McEwen 2015). As an example of the latter, capacity has been defined as the potential of an organisation resulting from the combination of available resources given by its context, such as human resources or financial resources (Homsy and Warner 2015; George and Reed 2016).

Capacity as action-oriented ability

In the second approach, authors refer to capacity in more action-oriented terms, such as when considering that the capacity of a public sector organisation in sustainability transitions is essentially related to their own ability to enact action (Förster, Downsborough et al. 2021). Hence, capacity is less about the potential according to an organisation’s situated position and the availability of resources, and more about the organisation’s own abilities to experiment, learn, and adapt when taking action (Borgström 2019). This approach focuses on the organisation’s ability to connect, to combine exploitation and exploration (their ambidexterity), to improve routines or services, or to learn (Gieske, van Buuren et al. 2016; Fiorino 2001). It also includes abilities related to experimenting (Borgström 2019), stewarding, unlocking, and orchestrating (Pedde, Kok et al. 2019). This approach contains an important reflexive element, as capacity also refers to the organisation’s ability for self-assessing and further developing its abilities (Meijer 2018), and its ability to actualise goals and strategies (Dent 2017). This approach implicitly suggests a dynamic perspective, as the capacity of an organisation may accrue or diminish over time, depending on the extent to which deliberate and continuous efforts inside the organisation are made to sustain and develop the organisation’s own abilities.

Capacity and capability

For its part, the notion of *capability* appears less frequently in the literature reviewed, but it has been gaining pace recently. Most scholars using the term *capability* have drawn from the strategic management literature in business studies to define the concept (Stenzel and Frenzel 2008; Lieberherr and Truffer 2015; Hartman, Gliedt et al. 2017; van Welie, Truffer et al. 2019; Mousavi and Bossink 2020; Zapata-Cantu and González 2021). The influential notion of *dynamic capabilities* was developed in the mid-1990s in the field of strategic management to study private firm’s differential ability to create and capture value in markets with rapid technological change (Teece, Pisano et al. 1997). In the context of sustainability transitions, Lieberherr and Truffer (2015), for example, have studied the dynamic capabilities of public utilities and defined those capabilities “as organisational and strategic routines that enable organisations to create, evolve, and recombine resources (ranging from physical assets to competences such as specific abilities) to generate new “value-creating” strategies and even change the market” (ibid, 103). It is worth noting that the literature on capabilities focuses on the abilities of the organisations, as routines and practices to engage in action adapting to a rapidly changing (market) context. For this reason, this literature sees capacity as organisations’ action developing their own abilities’ (Stenzel and Frenzel 2008); denoting an action-oriented approach to capacity, rather than the resource-based oriented approach mentioned earlier).

Combining approaches

Our review shows that few recent publications have started to use both concepts (capacity and capability) following the two approaches above (resource-based potential, and action-oriented ability). They tend to conceptualise capability as an organisation’s ability to perform certain diversified tasks. Hence, the capacity of a public sector organisation is seen “as a **set of resources** and **capabilities** central to perform policy functions” (Förster, Downsborough et al. 2021, 215). Thus, resources are different from capabilities, as capabilities refer to abilities in the form of practices, routines, and processes inside the organisation. “Investigating capacities means to unveil how actors are able or not to make use of their resources to achieve a certain outcome. An agent’s capacity is not necessarily equivalent to its resources. This is an important aspect because resources as such are not enough to foster or prevent policy implementation. Actors need to be able to utilise their resources such as trained staff, finance, or information” (Marquardt 2017, 232).

Following from this, we can see that capacity is about the interaction between resources (as situated assets internal and external to the organisation) and abilities (as the routines, practices, and procedures in the organisation). Fig. 2 shows this.

4. 研究发现：文献中的关键见解

上述文献综述提出了两个关键见解：一是关于能力的不同方法，二是关于变革能力中的“变革性”。

4.1. 关于能力的关键见解

我们观察到，部分综述文献使用了术语能力（*capacity*），而另一些则使用了能力（*capability*）；有些文献泛指能力（如制度能力），而另一些则涉及能力的具体维度（如分析能力）。在本小节中，我们将考察这些术语如何被概念化，识别其总体背景，最重要的是，它们之间如何相互关联。

能力（*Capacity*）是文献中使用最广泛的术语，我们可以识别出至少两种方法：能力作为基于资源的潜力，以及能力作为行动导向的能力。

能力作为基于资源的潜力

在第一种方法中，作者将能力定义为潜力或行动空间，即有待实现的机会（Olazabal and Pascual 2015），通常与资源相关（George and Reed 2016）。能力被视为组织在其给定资源下的总体潜力，即其运营许可，而资源通常由其制度定位/在特定背景下的处境所定义（Pelling, High et al. 2008）。因此，从这个意义上说，公共部门组织的能力指的是其根据所处资源拥有的总体潜力——例如政治授权、可用财政资源、合法性或人力资源——从而允许自主行动（Royles and McEwen 2015）。作为后的一个例子，能力被定义为组织由其背景给定的可用资源（如人力资源或财政资源）组合所产生的潜力（Homsy and Warner 2015; George and Reed 2016）。

作为行动导向的能力

在第二种方法中，作者以更具行动导向的术语来指代能力，例如，当考虑公共部门组织在可持续转型中的能力本质上与其自身采取行动的能力相关时（Förster, Downsborough et al. 2021）。因此，能力较少关乎组织根据其定位和资源可用性所拥有的潜力，更多关乎组织在行动时实验、学习和适应的自身能力（Borgström 2019）。这种方法专注于组织连接、结合开发与探索（双元性）、改进常规或服务以及学习的能力（Gieske, van Buuren et al. 2016; Fiorino 2001）。它还包括与实验（Borgström 2019）、管理、解锁和编排相关的能力（Pedde, Kok et al. 2019）。这种方法包含一个重要的反思元素，因为能力还指组织自我评估和进一步发展其能力的的能力（Meijer 2018），以及实现目标和战略的能力（Dent 2017）。这种方法隐含地提出了一种动态视角，因为组织的能力会随时间增减，这取决于组织内部是否为维持和发展自身能力而做出持续且故意的努力。

能力与能力

就其本身而言，能力这一概念在所综述文献中出现频率较低，但近年来越来越受关注。大多数学者使用能力一词时，借鉴了商业研究中战略管理文献来定义该概念（Stenzel 和 Frenzel 2008; Lieberherr 和 Truffer 2015; Hartman, Gliedt 等 2017; van Welie, Truffer 等 2019; Mousavi 和 Bossink 2020; Zapata-Cantu 和 González 2021）。动态能力这一影响深远的概念于 1990 年代中期在战略管理领域提出，用于研究私营企业在技术快速变化的市场中创造和获取价值的差异化能力（Teece, Pisano 等 1997）。在可持续转型背景下，Lieberherr 和 Truffer（2015）研究了公共事业部门的动态能力，并将其定义为“使组织能够创造、演化及重组资源（从实物资产到特定技能等能力）以生成新的‘价值创造’战略甚至改变市场的组织和战略惯例”（同上，103）。值得注意的是，关于能力的文献聚焦于组织的能力，即适应快速变化（市场）环境而采取行动的惯性和实践。因此，该文献将能力视为组织发展自身能力的行动（Stenzel 和 Frenzel 2008），表明能力是一种行动导向而非前述的资源导向方法。

结合不同方法

我们的综述表明，少数近期出版物已开始同时使用这两个概念（capacity 和 capability），遵循上述两种方法（基于资源的潜力与面向行动的能力）。它们倾向于将 capability 定义为一个组织执行特定多样化任务的能力。因此，公共部门组织的 capacity 被视为“一组资源和**capabilities**，这些对于履行政策职能至关重要”（Förster, Downsborough et al. 2021, 215）。因此，资源不同于 capabilities，因为 capabilities 指组织内部以实践、惯例和流程形式存在的能力。“研究 capacities 意味着揭示行动者如何能够或不能利用其资源来实现特定结果。一个行动者的 capacity 并不必然等同于其资源。这是一个重要方面，因为资源本身并不足以促进或阻碍政策实施。行动者需要能够利用其资源，如训练有素的人员、资金或信息”（Marquardt 2017, 232）。

由此可以看出，能力关乎资源（作为组织内外部的既有资产）与技能（作为组织中的常规、实践和程序）之间的相互作用。图2展示了这一点。

4.2. Key insights about transformativeness in transformative capacity

The previous key insights refer to capacity in general terms. We look now to how the literature sees *transformative change*. The first aspect to examine is the degree of change in transformative change. We start with the observation that there is a widespread understanding in the literature that sustainability transitions require deep change from current unsustainable modes of production and consumption (Isaksson and Hagbert 2020). Halfway solutions will not redress the current climate and biodiversity crisis. Aware of the path-breaking, deep change that is needed, the literature reflects a general understanding that this might not happen as an immediate or sudden upheaval of existing institutions and systemic features, but as a process characterised by the determination (and acceleration) towards path-breaking change lead by the changing roles of change agents (Avelino and Wittmayer 2016; Hagbert and Malmqvist 2019). The second aspect is whether the literature refers to transformative change as a process or as an outcome. We observe that both are present. Part of the literature focuses on it as a process, in terms of various pathways towards path-breaking change (Hölscher 2020; Borgström 2019); whereas another part of the literature focuses mostly on transformative change as a (policy) outcome (Homsy and Warner 2015; Kim and Yoon 2018). It is relevant to point that, in both cases, the transformative capacity of organisations is widely seen as a factor in the process of path-breaking change, and as a factor (or variable) for explaining change as an outcome. From the organisational perspective, transformative capacity is the capacity of change agents (including PSOs) in their exercise of path-breaking change agency through directionality, vision, tactics, and imagination, with (eventually) transformative (policy) outcomes (Förster, Downsborough et al. 2021). We observe as well that the literature does not treat the transformative capacity of an organisation as a single feature. Instead, it treats it as a set of various elements that the organisation puts together strategically to enact and work towards the transformative goal (Hagbert and Malmqvist 2019; Gieske, van Buuren et al. 2016; Anadon, Chan et al. 2016; Weber and Rohracher 2012). In other words, transformative capacity is a combination of various elements, relating to the various roles that the organisation defines and assumes when working purposefully to enact change agency, in the specific context in which the organisation is embedded. This notion of *the purposeful work of change agents* brings us to the topic of roles. We deal with it in the next section, when we conceptualise the constitutive elements of PSOs’ transformative capacity.

5. Conceptualising roles, resources, and abilities as constitutive elements of PSOs transformative capacity

The insights above offer us the opportunity to engage in a conceptualisation of some constitutive elements, bringing forward a higher-order framework that focus explicitly on PSOs pursuing system-wide transformations. To do this, we adopt an organisational perspective, understanding that PSOs are embedded in complex, multi-level contexts that vary across countries, geographies, time, and traditions. We take the perspective that each public sector organisation is a distinct and unique organisation with its own approach and idiosyncrasy. The findings and key insights allow us to identify three core constitutive elements. The first element is the various roles that PSOs play in their exercise of purposeful tasks when exercising agency towards transformation. The second element is the various situational resources that can be mobilised by PSOs to enact their transformative roles. The third element is the set of abilities, which are the routines, practices, and procedures inside the organisation when exercising transformative agency.

5.1. Roles

The literature reviewed tends to underline that roles are performed in interaction with other actors, and that actors might change their roles through time. It mentions as well, without digging deeper, that roles might be ascribed or constructed, and that organisations might (or might not) assume them (Avelino and Wittmayer 2016). Part of the literature sees PSOs mainly as system-maintaining actors (Richter and Mundaca 2015; Bowen and Panagiotopoulos 2020), whereas others observe that PSOs are actively implementing the transformative change defined by political and collective climate/green targets (Anadon, Chan et al. 2016; Fischer and Newig 2016). The literature points at different transformative roles, but does not provide a wide and encompassing view

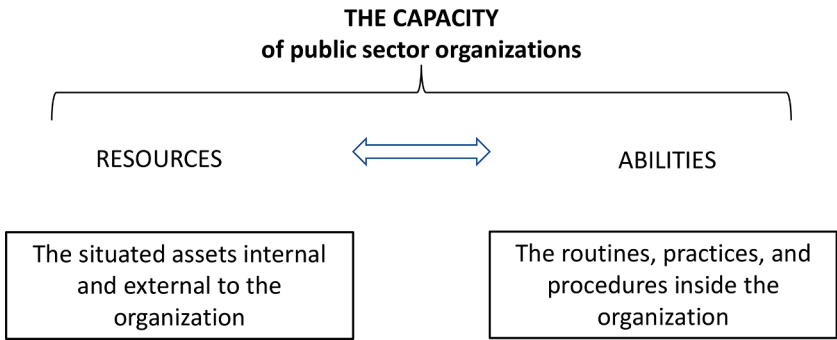


Fig. 2. The capacity of PSOs as the interaction between situated resources and abilities of the organisation.

4.2. 关于转型能力中转型性的关键见解

之前的核心见解从一般意义上讨论了能力。现在我们要看文献如何理解变革性转型。首先要考察的是转型变革中的变化程度。我们注意到，文献中普遍认为，可持续转型需要从当前不可持续的生产和消费模式进行深度变革 (Isaksson and Hagbert 2020)。折中方案无法解决当前的气候和生物多样性危机。鉴于需要开创性的深度变革，文献普遍认为，这可能不会立即或突然颠覆现有制度和系统特征，而是一个由变革者角色变化主导的、逐步（并加速）走向开创性变革的过程 (Avelino and Wittmayer 2016; Hagbert and Malmqvist 2019)。

第二个方面是文献中将转型性变革视为过程还是结果。我们发现两者都存在。部分文献将其视为过程，即走向开创性变革的各种路径 (Hölscher 2020; Borgström 2019)；而另一部分文献则主要将转型性变革视为（政策）结果 (Homsy and Warner 2015; Kim and Yoon 2018)。值得注意的是，在这两种情况下，组织的转型能力都被广泛视为开创性变革过程中的一个因素，以及解释变革作为结果的一个因素（或变量）。从组织角度来看，转型能力是变革者（包括公共部门组织）通过方向性、愿景、策略和想象力行使开创性变革的能力，最终产生转型性（政策）结果 (Förster, Downsborough et al. 2021)。

我们还观察到，文献并未将组织的变革能力视为单一特征，而是将其视为组织为实现变革目标而战略性地组合的一系列不同要素 (Hagbert and Malmqvist 2019; Gieske, van Buuren et al. 2016; Anadon, Chan et al. 2016; Weber and Rohracher 2012)。换言之，变革能力是多种要素的组合，与组织在特定背景下为有目的地推动变革而定义和扮演的各种角色相关。这种变革推动者有目的的工作概念引出了角色的话题。我们在下一节中讨论这一点，并在其中将PSO’ 变革能力的构成要素概念化。

5. 概念化 角色、资源和能力作为公共部门组织变革能力的构成要素 y

上述见解为我们提供了对某些构成要素进行概念化的机会，从而提出了一个明确关注追求系统范围转型的公共部门组织的高阶框架。为此，我们采用组织视角，理解公共部门组织嵌入在不同国家、地理、时间和传统的复杂多层次背景中。我们认为每个公共部门组织都是一个独特且各具特色的组织，有其自身的方法和特质。

这些发现和关键见解使我们能够识别出三个核心构成要素。第一个要素是公共部门组织在推动转型时，在履行有目的的任务过程中扮演的各种角色。第二个要素是公共部门组织可以调动以发挥其转型作用的各种情境资源。第三个要素是一组能力，即组织内部在行使转型能动性时的惯例、实践和程序。

5.1. 角色

所评述的文献倾向于强调，角色是在与其他行动者的互动中扮演的，并且行动者可能随时间改变其角色。文献也提到（但未深入探讨）角色可能是被赋予的或建构的，而组织可能（也可能不）承担这些角色 (Avelino and Wittmayer 2016)。部分文献将公共部门组织 (PSOs) 主要视为系统维护者 (Richter and Mundaca 2015; Bowen and Panagiotopoulos 2020)，而其他文献则观察到 PSOs正在积极实施由政治和集体气候/绿色目标所定义的变革性转变 (Anadon, Chan et al. 2016; Fischer and Newig 2016)。文献指出了不同的变革性角色，但并未提供对这些角色的广泛而全面的看法

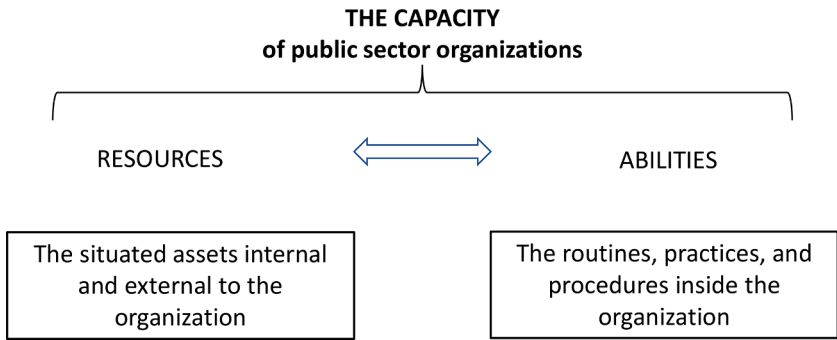


图2. 公共部门组织的能力是情境资源与组织能力之间的互动。

on them, nor a definition of what those roles are. Below we summarize the roles identified by the literature and provide a definition of role as institutional work.

PSOs might act as *promoters and champions* of visions of change (De Laurentis and Pearson 2021), leveraging the collective imagination of possible futures (Moore and Milkoreit 2020).

They might as well assume the role as *designers of green transformation*, or facilitators, for example, when local administrations actively plan the transformation of the urban context defining ambitious green local plans and restructuring municipal forms of intervention (Frantzeskaki and Tilie 2014), as well as working to overcome existing legal barriers (for example in the permits for installing solar panels) (Quitza, Hoffmann et al. 2012).

Within the conflictual dimensions of transitions, PSOs might take the role as *mediators in conflicts*. The literature also notes that the dialogue or controversy between change-oriented actors and incumbents (or positively and negatively affected parties) matters for how PSOs define their roles (George and Reed 2016; Meijer 2018; Hessevik 2021), as specific public agencies might work towards accommodating various alternative perspectives, addressing value- and distributional conflicts (Kay and Ackrill 2012).

PSOs might also act as active *partner-builders*, brokering to build alliances, collaboration and specific partnerships with heterogeneous actors (George and Reed 2016; Grotenbreg and van Buuren 2018). The lack of performing such a role as partner-builder might lead to green transition failures (Huang 2021).

Enablers of wide societal participation is an important role as well, as PSOs create frameworks and channels for the participation of social actors, seeking to engage them as co-producers of solutions (Wolfram 2018; Novalia, Rogers et al. 2020).

Overall, we see the literature acknowledging the importance of different roles. However, the individual publications tend to focus only on one, or on few of them, typically disregarding that the same public organisations might perform several roles at once.

Likewise, there is a general trend in the literature to disregard the possible dilemmas or paradoxes of performing transformative roles in complex contexts, or the variations of possible roles according to the type of sustainability transition and its governance modes (Borrás and Edler 2020).

When dealing with transformative capacity, we define roles as the variety of specific, purposeful tasks that an organisation performs while enacting institutional work towards transformation in sustainability transitions. We note the current debate about the role of the state in sustainability transitions and transformative processes, and we embrace the variety of roles that the state and its PSOs might assume (Mazzucato 2015; Borrás and Edler 2020). We also acknowledge that those diverse roles have different purposeful character, and therefore might follow the guidance provided by the literature on institutional work. Institutional work is “the purposive action of individuals and organisations aimed at creating, maintaining and disrupting institutions” (Lawrence and Suddaby 2006, 215), an approach that is particularly relevant for studying how organisations assume specific roles when engaging in sustainability transitions (Beunen and Patterson 2019; Kivimaa, Boon et al. 2019). Hence, we consider that PSOs might assume various roles that aim at disrupting, creating, and/or maintaining/stabilising institutions, which taken together work towards path-breaking change. PSOs might do that according to the goals given to them in explicit political targets, and/or according to their own organisational ethos and initiative.

5.2. Resources

The reviewed literature repeatedly mentions resources in relation to transformative capacity. Often, there is no definition of resources, and they are mentioned more than they are analysed. Below we summarize the resources most often mentioned and provide a definition of resources.

Human resources, such as staff, manpower, personnel, etc.; and *financial resources* (budget, funding, etc.), are widely seen as important resources (Spyridaki, Kleanthis et al. 2020).

Data and information are also important resources for PSOs’ capacity for sustainability transitions (Bowen and Panagiotopoulos 2020; Salvador and Sancho 2021).

Physical resources, such as natural resources, infrastructure, geographical location and geological features are also important, defining the locational, physical and materiality context for PSOs action (De Laurentis and Pearson 2021; Marquardt 2017).

Political mandates are another crucial type of resource. Mandates can be *explicit*, like the legal-constitutional aspects of PSOs’ own remit, or concrete political mandates assigned to the PSO - like concrete climate targets at national or local levelsⁱ (Royles and McEwen 2015; Beck, Jasanoff et al. 2021). But mandates can also be *implicit*, for example, from international policy strategies, which are clear reference points for PSOs’ action. Whether explicit and/or implicit, political mandates are highly relevant resources because they are the PSOs’ ‘license to operate’ towards transformative change.

Network resources, are the PSO’s number and type of contacts, access to influential decision-makers, and/or relations with private actors. Networks are recurrently seen as relevant resources (Grotenbreg and van Buuren 2018; Hölscher 2020).

Cultural resources in terms of public administration/bureaucratic culture, and general sustainability-culture and awareness in the society, are also mentioned in the literature (Larson, Lauber et al. 2017; Avelino and Wittmayer 2016).

Legitimacy resources based on the PSO’s reputation, social acceptance, social authority, trustworthiness, and general civic traditions, are also relevant (Homsy and Warner 2015) (Pelling, High et al. 2008; Boer 2013); but tend to be mentioned less frequently.

Based on the above, we define resources as the set of material or immaterial situated assets that are available (and/or perceived to

ⁱ A specific example of an explicit mandate is the delegation of political competencies for sustainability transitions from national to lower levels of government (local or regional authorities).

，也未定义这些角色是什么。下面我们总结了文献中识别的角色，并将角色定义为制度工作。

公共部门组织可能扮演变革愿景的推动者和倡导者（De Laurentis and Pearson 2021），利用对未来可能的集体想象（Moore and Milkoreit 2020）。

它们也可能承担绿色转型设计者或促进者的角色，例如，当地方行政部门积极规划城市转型，制定雄心勃勃的绿色地方计划并重组市政干预形式时（Frantzeskaki and Tilie 2014），同时努力克服现有法律障碍（例如安装太阳能电池板的许可）（Quitza, Hoffmann et al. 2012）。

在转型的冲突维度中，公共部门组织可能扮演冲突调解者的角色。文献还指出，变革导向的行动者与现有参与者（或积极与消极影响的各方）之间的对话或争议，影响着公共部门组织如何定义其角色（George and Reed 2016; Meijer 2018; Hessevik 2021），因为特定的公共机构可能致力于容纳各种不同的观点，处理价值和分配冲突（Kay and Ackrill 2012）。

公共部门组织也可能充当积极的合作伙伴构建者，通过中介作用建立联盟、协作以及与异质行动者的特定伙伴关系（George and Reed 2016; Grotenbreg and van Buuren 2018）。缺乏履行这种合作伙伴构建者的角色可能导致绿色转型失败（Huang 2021）。

广泛社会参与的促进者也是一个重要角色，因为公共部门组织为社会行动者的参与创建了框架和渠道，试图让他们成为解决方案的共同生产者（Wolfram 2018; Novalia, Rogers et al. 2020）。

总体而言，我们看到文献承认了不同角色的重要性。然而，个别出版物往往只关注其中一个或少数几个角色，通常忽视了同一公共组织可能同时扮演多个角色的事实。

同样，文献中普遍存在一种趋势，即忽视在复杂情境中扮演变革角色的可能困境或悖论，或忽视根据可持续转型类型及其治理模式可能产生的角色变化（Borrás and Edler 2020）。

在处理转型能力时，我们将角色定义为一个组织在执行制度工作以实现可持续发展转型过程中所执行的各种具体、有目的的任务。我们注意到当前关于国家在可持续发展转型和转型过程中作用的讨论，并接受国家及其公共部门组织可能承担的各种角色（Mazzucato 2015; Borrás and Edler 2020）。我们还认识到，这些不同的角色具有不同的目的性，因此可能遵循制度工作文献提供的指导。制度工作是指“个人和组织旨在创造、维持和破坏制度的有目的行动”（Lawrence and Suddaby 2006，第215页），这种方法对于研究组织在参与可持续发展转型时如何承担特定角色特别相关（Beunen and Patterson 2019; Kivimaa, Boon et al. 2019）。因此，我们认为公共部门组织可能承担各种角色，旨在破坏、创造和/或维持/稳定制度，这些角色共同致力于实现突破性变革。公共部门组织可能根据明确的政策目标赋予他们的目标，和/或根据他们自身的组织精神和主动性来做到这一点。

5.2. 资源

所回顾的文献中反复提及资源与转型能力的关系。通常，资源没有明确定义，它们被提及的次数多于被分析的次数。下面我们总结最常提及的资源，并给出资源的定义。

人力资源，如员工、人力、人员等；以及财力资源（预算、资金等），被广泛视为重要资源（Spyridaki, Kleanthis et al. 2020）。

数据和信息也是公共部门组织'可持续转型能力的重要资源（Bowen and Panagiotopoulos 2020; Salvador and Sancho 2021）。

物质资源，如自然资源、基础设施、地理位置和地质特征，也很重要，它们定义了公共部门组织行动的地点、物理和物质上下文（De Laurentis and Pearson 2021; Marquardt 2017）。

政治授权是另一种关键资源。授权可以是显性的，如公共部门组织'自身职责的法律宪法方面，或分配给公共部门组织的具体政治授权——例如国家或地方层面的具体气候目标ⁱ（Royles and McEwen 2015; Beck, Jasanoff et al. 2021）。但授权也可以是隐性的，例如来自国际政策战略，这些是公共部门组织'行动的明确参考点。无论是显性和/或隐性，政治授权都是高度相关的资源，因为它们公共部门组织'“运营许可”，朝着变革性改变的方向。

网络资源是公共部门组织的联系数量和类型、接触有影响力的决策者、以及/或与私人行为者的关系。网络被视为重要的资源（Grotenbreg and van Buuren 2018; Hölscher 2020）。

文化资源在公共行政/官僚文化、社会中的总体可持续性文化和意识方面，文献中也有所提及（Larson, Lauber et al. 2017; Avelino and Wittmayer 2016）。

合法性资源基于公共部门组织的声誉、社会接受度、社会权威、可信度以及一般公民传统，也很重要（Homsy and Warner 2015）（Pelling, High et al. 2008; Boer 2013）；但提及频率较低。

基于上述，我们将资源定义为可用（和/或感知为）的一组物质或非物质情境性资产，

ⁱ 一个明确授权的具体例子是将可持续转型的政治能力从国家下放到下级政府（地方或区域当局）。

be available) to a public sector organisation, either internally or externally to the organisation. The situated nature of resources is important because PSOs are not free to define the level or the availability of their resources, given the political and located nature of resource allocation. This contrasts with non-state, private organisations that have more leeway in those terms (e.g. firms, civil society organisations, NGOs, etc.). Often, PSOs operate under considerable resource scarcity due to efficiency-focused political goals, that have reduced public sector resources in significant ways over the past decades. Likewise, the resources of PSOs vary according to their situated position in the socio-political context: it matters whether they are a large and central regulatory agency or a small executive agency, a wealthy municipality or a socio-economically deprived municipality. This however does not mean that resources are fixed; they might change through time, as the organisation might make efforts to get access to new resources and reinterpret existing ones.

5.3. Abilities

The third element of the transformative capacity of PSOs in sustainable transitions found in the literature is the set of PSO abilities. Our review shows that the literature focuses on various abilities at the organisational level, clustering around four distinct sets: analytical, operational, coordination, and learning/reflection abilities.

Analytical abilities refer generally to exploring, studying, sense-making, and interpreting. It is “the ability of organisations to produce valuable research and analysis on topics of their choosing” (Oliphant and Howlett 2010, 439), gathering information and assessing it (Kattel 2022), processing and developing new ideas and visions about the future that allow new forms of public innovation (Meijer 2018).

Operational abilities are those frequently described as the managerial and administrative abilities to deliver in practical terms: executing and exploiting the possibilities in actionable ways (Spyridaki, Kleanthis et al. 2020); as well as monitoring, verifying, and securing (legal) enforcement (Borzel and Buzogany 2019; Bowen and Panagiotopoulos 2020; He and Hultman 2021; Tao and Mah 2009; Li and Chan 2009; Schmid and Bornemann 2019); and shaping and embedding new routines (Kattel 2022).

Coordination abilities are typically seen as routines for collaborating and connecting with other organisations (Mu, Jia et al. 2018; Bissix and Rees 2001). They refer to the ability of PSOs to stimulate and orchestrate multi-stakeholder participation, communication, and social engagement (George and Reed 2016; Hessevik 2021; Strasser, de Kraker et al. 2019).

Learning and reflexivity abilities are a public organisation’s ability to develop and incorporate new understandings and adjust its own action accordingly, through adaptation and re-consideration (Monstadt and Wolff 2015). Sometimes learning and reflexivity abilities are subsumed under analytical abilities; other times, the literature considers them distinct.

Having considered this discussion in the literature, we define abilities here as the set of routines, practices, and procedures of the organisation in the enactment of the organisation’s change agency. Naturally, those routines have to do with the various ways in which the organisation mobilises and utilises its situated resources as it enacts its defined roles.

Table 1 summarizes the conceptualization or roles, resources and abilities.

6. Towards a holistic framework of transformative capacity

Understanding that the transformative capacity of a PSO is its capacity to enact path-breaking change agency (seeking to overcome heavy technological lock-ins and barriers in system transformation), we have three theoretical considerations. First, that the transformative capacity of a PSO is not static, but the result of the continuous interaction between the roles, resources, and abilities as three constitutive elements. Thus, the question we seek to answer is how they interact. Second, that a PSO’s transformative capacity co-

Table 1
Conceptualizing Roles, Resources and Abilities.

Constitutive element	Types according to the literature
Roles	Promoters and champions of visions Designers of green transformations Mediators in conflicts Partner-builders Enablers of wide societal participation (Purposeful roles maintaining, changing and/or disrupting institutions)
Resources	Human resources Financial resources Data and information resources Physical/natural resources Political mandates Network resources Cultural resources Legitimacy resources
Abilities	Analytical abilities (exploring, studying, investigating...) Operational abilities (exploiting, managing, executing...) Coordination abilities (collaborating, connecting, communicating...) Learning and reflectivity abilities (incorporating new understandings, adjusting...)

Own Source

公共部门组织（内部或外部）可以获取这些资源。资源的所处情境很重要，因为公共部门组织不能自由定义其资源的水平或可用性，这取决于资源分配的政治性和地域性。这与非国家、私人组织（如企业、公民社会组织、非政府组织等）形成对比，后者在这些方面有更多自由度。由于过去几十年注重效率的政治目标大幅削减了公共部门资源，公共部门组织常常在资源严重匮乏的情况下运作。同样，公共部门组织的资源因其在社会政治环境中所处位置而异：它们是大而核心的监管机构还是小型执行机构、富裕城市还是社会经济欠发达城市，都会产生影响。但这并不意味着资源是固定的；随着时间的推移，资源可能会发生变化，因为组织可能会努力获取新资源并重新解释现有资源。

5.3. 能力

文献中发现的公共部门组织在可持续转型中变革能力的第三个要素是一系列公共部门组织能力。我们的综述表明，文献关注组织层面的各种能力，集中在四个不同的集合：分析能力、操作能力、协调能力和学习/反思能力。

分析能力通常指探索、研究、意义建构和解释。它是“组织就其所选主题产生有价值研究和分析的能力” (Oliphant and Howlett 2010, 439)，收集信息并评估它 (Kattel 2022)，处理和开发关于未来的新想法和愿景，从而允许新的公共创新形式 (Meijer 2018)。

操作能力通常被描述为在实际交付中的管理和行政能力：以可操作的方式执行和利用可能性 (Spyridaki, Kleanthis et al. 2020)；以及监督、验证和确保（法律）执行 (Borzel and Buzogany 2019; Bowen and Panagiotopoulos 2020; He and Hultman 2021; Tao and Mah 2009; Li and Chan 2009; Schmid and Bornemann 2019)；以及塑造和嵌入新惯例 (Kattel 2022)。

协调能力通常被视为与其他组织协作和联系的常规 (Mu, Jia et al. 2018; Bissix and Rees 2001)。它们指公共部门组织激发和协调多方利益相关者参与、沟通和社会参与的能力 (George and Reed 2016; Hessevik 2021; Strasser, de Kraker et al. 2019)。

学习和反思能力是公共组织通过适应和反思 (Monstadt and Wolff 2015) 来发展并整合新认识、相应调整自身行动的能力。有时，学习和反思能力被归入分析能力之下；其他时候，文献认为它们截然不同。

在考虑了文献中的这一讨论后，我们将能力定义为组织在其变革代理人角色实施过程中的一套常规、实践和程序。自然，这些常规与组织在实施其既定角色时，如何动员和利用其情境资源的各种方式有关。

表1概括了角色、资源和能力的概念化。

6. 迈向整体性的转型能力框架

理解公共部门组织的变革能力是其实施突破性变革行动的能力（旨在克服系统转型中的严重技术锁定和障碍），我们有三点理论思考。首先，公共部门组织的变革能力并非静态，而是角色、资源和能力三个构成要素持续互动的结果。因此，我们试图回答的问题是它们如何互动。其次，公共部门组织的变革能力与

表1角色、资源和能力的概念化

构成要素	文献中的类型
角色	愿景的推动者和倡导者 绿色转型的设计师 冲突中的调解者 合作伙伴的构建者 广泛社会参与的促成者 （有目的地维持、改变和/或打破制度的角色）
资源	人力资源 财政资源 数据和信息资源 物理/自然资源 政治授权 网络资源 文化资源 合法性资源
能力	分析能力（探索、研究、调查……） 操作能力（利用、管理、执行……） 协调能力（协作、连接、沟通……） 学习与反思能力（吸收新理解、调整……）

自有来源

evolves through time with the phases that sustainability transitions’ pathways might go through. And third, that a PSO might be able to build its transformative capacity by a gradual expansion of the PSO’s constitutive elements.

6.1. Transformative capacity as interactions between roles, resources, and abilities

Beginning with the first theoretical consideration, the transformative capacity of a PSO is defined by how it enacts the interactions between the three constituent elements, creating specific interactions that carry transformative change agency. In other words, the way it enacts the interactions is not a static feature.

As we saw earlier in the context of transformative capacity, we understand roles as those that a PSO actively assumes when pursuing transformative change. These are enacted in the interlinkage with its resources and abilities. Hence, a PSO might have a relatively good availability of resources (financial, human, data, political mandate, etc.) and well-established routines, but without a clear self-definition of roles and specific tasks that purposely seek directional change, those resources and abilities might be to no avail for sustainability transitions. This is because sustainability transitions require purposeful action to overcome heavy technological lock-ins, incumbents’ opposition, and other important hurdles. Likewise, some PSOs might have reduced resources and less developed abilities; yet, a clear and determined purposeful definition of roles might carry on transformative change agency by mobilising these scarce resources and by (rapidly) re-configuring the routines in PSO’s everyday work. This consideration suggests that roles are necessary conditions, while resources and abilities are sufficient conditions for a PSO’s transformative capacity. This is a hypothesis that would need to be empirically examined in the future.

It is important to admit that some PSOs might not have any transformative capacity at all if their roles, resources, and abilities are severely limited or never unfolded. Nevertheless, this should not lead us to automatically assume that a PSO’s transformative capacity is static, as it is the result of the continuous interactions between the PSO’s roles, resources, and abilities; and might change through time. Nor should we assume that transformative capacity takes only one possible form, as similar PSOs might enact transformative capacity in different ways according to variations of these interactions.

How might we conceptualise that diversity of interactions? One way of considering it theoretically would be to take the starting point from a PSO’s roles, which we hypothesise are necessary conditions for a PSO’s transformative capacity. One PSO might combine various roles which, taken together, seek to enact transformative change. For example, a local government (the public administration implementing city council political decisions) engaged in the acceleration of renewable energy sources might simultaneously enact roles as lead user, initiator, facilitator, and negotiator (all of them combined are the PSO’s institutional work towards transformative change). As a *lead user*, it can use its control over the local utility to actively push to phase-in renewable energy sources and phase out fossil sources in the production of district heating. As an *initiator*, it can install solar panels on the roofs of its own municipal schools and elderly homes. As a *facilitator*, it can create new and foreseeable approval procedures while stimulating third parties (e.g. citizens, local firms, energy developers) to invest in renewable energy locally. As a *negotiator*, it can actively lobby the national level public administration to allow for a specific interpretation/adaptation of national regulations that would ease these initiatives. The intensity and breadth of a PSO’s roles, combined, might have path-breaking effects locally and perhaps also nationally.

Clearly, the enacting of each of these roles is linked to the PSO’s resources and abilities. To be sure, local governments are the administrative branch of politically elected city councils, and for that reason, the social legitimacy behind the political mandate associated with the local green transition goals is an important resource for a PSO, though it is not always determinant. By contrast, in situations with low social legitimacy or a high level of conflict around those local goals, a PSO might have limited socio-political resources to mobilise. Financial and human resources are also important, mainly when a PSO acts as *initiator* because actions like installing solar panels on the roofs of municipal buildings are financially costly and require staffing. Without those resources, a PSO might not be able to enact that role. Similarly important is a PSO’s operational ability because the operation of those solar panels requires specific technical verification and maintenance procedures which are new to the PSO.

Less demanding in terms of financial resources is the role of *facilitator* because these are mainly third-party investments. However, that role requires PSO data and network resources, as it needs access to relevant information and expertise, allowing municipal planners to assess and eventually give (or not) permission to install those third-party investments. Likewise, this role requires good coordination abilities across various PSO units, aligning administrative criteria and legal-technical interpretation/development of local rules, as well as managing third-party expectations around key legal enforcement aspects associated with those permissions.

Overall, the larger the number of transformative roles a PSO assumes, the more demanding the types of resources it needs to mobilise and the types of abilities it needs to develop. This brings us to two further considerations, the process of transformation and the building of transformative capacity. We deal with them in the next two sub-sections.

6.2. The co-evolution between transformative pathways and pso roles, resources, and abilities

We revert now to questions about the process of transformation. Since the seminal work of Geels and Schot about different pathways to sustainability transitions, the literature has largely acknowledged that transformative change is not about one single type of pathway (Geels and Schot 2007). Moreover, alternative pathways to achieve the same endpoint differ in terms of lead actors and policy mixes, given the different timing and forms of barriers to overcome (Rogge, Pfluger et al. 2020). Deep transformations take time and accelerating them requires overcoming a significant number of barriers, not least imagining and finding technologies and solutions that do not yet exist. For this reason, the literature offers an endless list of possible barriers and drivers, all depending on the idiosyncrasies of specific transformative pathways, and specific needs for accelerating transitions.

This is related to the ‘sense of direction’ of transformative agency, and therefore of PSOs roles, resources and abilities. Hence, in the

可持续转型路径可能经历的阶段随时间共同演化。第三， 公共部门组织可能通过逐步扩展其构成要素来建立自身的变革能力。

6.1. 变革能力作为角色、资源和能力之间的互动

从第一个理论考量开始， 公共部门组织的变革能力取决于它如何实施三个构成要素之间的互动， 从而形成承载变革推动力的特定互动。换言之， 其实施互动的方式并非静态特征。

正如我们之前在变革能力的语境中看到的， 我们将角色理解为公共部门组织在追求变革时主动扮演的角色。这些角色通过与其资源和能力的相互关联而得以实施。因此， 一个公共部门组织可能拥有相对充足的资源（财务、人力、数据、政治授权等）和成熟的常规流程， 但如果没有明确的角色自我定义和专门寻求方向性变革的具体任务， 这些资源和能力可能对可持续转型毫无用处。这是因为可持续转型需要有针对性的行动来克服严重的技术锁定、 现有企业的抵制以及其他重大障碍。同样， 一些公共部门组织可能资源减少且能力较弱； 然而， 明确而坚定的目的性角色定义可以通过动员这些稀缺资源并（迅速）重新配置公共部门组织日常工作流程来推动变革。这一考量表明， 角色是必要条件， 而资源和能力是公共部门组织变革能力的充分条件。这是一个假设， 未来需要通过实证检验。

必须承认， 如果一些公共部门组织的角色、资源和能力受到严重限制或从未发挥， 它们可能根本不具备任何变革能力。然而， 这不应让我们自动认为公共部门组织的变革能力是静态的， 因为它是该组织角色、资源和能力之间持续互动的结果， 并且可能随时间变化。我们也不应假设变革能力只有一种可能的形式， 因为相似的公共部门组织可能根据这些互动的差异以不同方式展现变革能力。

我们如何概念化这种互动的多样性？ 理论上的一种方式， 从公共部门组织的角色出发， 我们假设这些角色是该组织变革能力的必要条件。一个公共部门组织可能结合多种角色， 共同寻求实现变革。例如， 参与加速可再生能源的地方政府（执行市议会政治决策的公共行政机构）可能同时扮演领先用户、发起者、推动者和谈判者的角色（所有这些角色共同构成该组织推动变革的制度工作）。作为领先用户， 它可以通过对地方公用事业的控制， 积极推动在区域供暖生产中引入可再生能源并淘汰化石能源。作为发起者， 它可以在自己的市政学校和养老院屋顶安装太阳能电池板。作为推动者， 它可以创建新的、可预见的审批程序， 同时激励第三方（如公民、当地企业、能源开发商）在当地投资可再生能源。作为谈判者， 它可以积极游说国家层面的公共行政机构， 允许对国家法规进行特定的解释或调整， 以简化这些举措。该组织角色的强度和广度相结合， 可能对地方甚至国家产生路径突破性影响。

显然， 每种角色的实施都与公共部门组织的资源和能力相关联。诚然， 地方政府是政治选举产生的市议会的行政部门， 因此， 与地方绿色转型目标相关的政治授权背后的社会合法性是公共部门组织的重要资源， 尽管它并不总是决定性的。相反， 在那些地方目标的社会合法性较低或冲突程度较高的情况下， 公共部门组织可能拥有有限的社会政治资源可供调动。财政和人力资源也很重要， 尤其是在公共部门组织扮演发起者角色时， 因为在市政建筑屋顶安装太阳能电池板等行动成本高昂且需要人员配置。缺乏这些资源， 公共部门组织可能无法履行该角色。同样重要的是公共部门组织的运营能力， 因为太阳能电池板的运行需要特定的技术验证和维护程序， 这对公共部门组织来说是全新的。

从财政资源角度看， 要求较低的是促进者角色， 因为这些主要是第三方投资。然而， 该角色需要公共部门组织的数据和网络资源， 因为它需要获取相关信息和专业知

识， 使城市规划者能够评估并最终允许（或不允许）安装这些第三方投资。同样， 该角色要求公共部门组织各部门之间具有良好的协调能力， 统一行政标准及地方法规的法律技术解释/制定， 并管理第三方在与这些许可相关的法律执行方面的期望。

6.2. 变革路径与公共部门组织角色、资源及能力的共同演进

现在我们回到关于变革过程的问题。自Geels和Schot关于可持续转型不同路径的开创性工作以来， 文献广泛承认变革性变化并非单一类型的路径（Geels and Schot 2007）。此外， 实现相同终点的替代路径在主要行动者和政策组合方面存在差异， 因为需要克服的障碍的时机和形式不同（Rogge, Pfluger et al. 2020）。深层转型需要时间， 加速转型需要克服大量障碍， 尤其是构想和寻找尚不存在的技术和解决方案。因此， 文献提供了无尽的障碍和驱动因素清单， 这些取决于特定变革路径的特殊性以及加速转型的具体需求。

这与变革性能动性的 ‘方向感’ 有关， 因此也涉及公共部门组织（PSOs）的角色、资源和能力。因此， 在

early phases of transformative change, a PSO might act as an initiator and lead user, as mentioned above. Yet PSOs are not always in the lead, as other pathways are led by societal or market actors. In such situations, a PSO’s transformative capacity roles are mainly to act as a facilitator (encouraging others to engage) and to prevent others (or themselves) from becoming barriers. The central position of some PSOs are crucial in this regard. For example, in the shipping sector, a sustainability transition involves private ship-owners’ replacement of fossil fuelled ships with new vessels that sail on zero-emission fuels. Maritime regulatory agencies are PSOs, which approve ships in order to ensure maritime safety. Without their approval, no zero-emission fuel ships ship can set sail, and the PSO is crucial for deployment of green technologies in the shipping sector. Existing regulations are attuned to fossil fuel technologies, and the legal-technical expertise of the maritime regulatory authorities is crucial in redefining the regulatory framework, acting as early negotiators at international and national regulatory levels. Once the framework is clarified and settled to enable new fuels, the agencies start approval work of each new ship. Maritime regulatory agency needs to mobilize new resources (such as new data and external expertise about new technologies), while developing its own internal analytical abilities to engage in approval work. In later phases of the transition, when the zero-emission ships are operating, the maritime agency verifies and monitors safety. This requires the agency to develop yet another set of operational abilities because these ships’ safe operations differ radically from fossil-fuelled ships. All in all, the transformative roles, resources, and abilities of the maritime agencies differ over the different phases of the transition pathway (Taudal Poulsen and Borrás).

Much of the literature underlines that transformative change is rarely a linear, conflict-free, or costless process. Depending on how and when the tensions unfold, a PSO might take various roles, engaging to minimise these tensions to avoid the transition getting delayed or derailed. In the early phases, a PSO might act as a negotiator between different opponents. It might also act as a promoter of specific approaches, actively endorsing one side in the contention. In both cases, whether acting as a negotiator or as a promoter of a specific side, a PSO needs to mobilise its own legitimacy resources (based on the credibility that it is working towards the public interest), and it must build a convincing narrative. This might not be easy at times. Likewise, it needs strong communication and coordination abilities, to manage that communication. Later, a PSO might also engage in a role as a mitigator, seeking to find ways of compensating the negative effects of transition. This might require a PSO to mobilise financial resources and operational abilities, for reimbursing/compensating them, as well as data resources and analytical abilities for evidence-based interventions about which societal groups are most affected and how to best compensate them.

All the above suggests there might be a co-evolution between the stages of the transition processes and the transformative capacity of a PSO, both changing through time. Whereas we have seen that it might change through time, we cannot assume that a PSO’s transformative capacity is automatically expanding and growing. This brings us to the last theoretical consideration of this article about building transformative capacity.

6.3. Building transformative capacity in public sector organisations

A third theoretical consideration is about building transformative capacity. We take as the starting point from PSO’s roles, assuming they are a necessary condition for a PSO’s transformative capacity in the enactment of purposeful change agency. Following this, we can then assume that building transformative capacity in a PSO is not an automatic process. To some extent, transformative capacity is built from inside the PSO itself, naturally in relation to the wider governance, and public administrative context in which the PSO operates. Hence, two further considerations might be relevant here: capacity-building as a process of gradual expansion of its constitutive elements; and capacity-building as a process related to public leadership.

Starting with the first, we can theorise that transformative capacity is built through an interactive process of expanding a PSO’s own constitutive elements, namely roles, resources, and abilities. Their interaction means they might (or might not) gradually support each other, expanding in terms of breadth and intensity. The notion of ambidexterity as exploiting and exploring abilities might be relevant to explore in this regard (Gieske, van Buuren et al. 2016). Building transformative capacity is not a PSO’s mere response to external socio-political demands in the mechanistic implementation of collective targets. To a certain degree it is a strategic move of the PSO (triggered by the PSO itself or by its principal – a ministry, etc.) to create a more powerful contextual positioning in terms of enacting transformative action. ‘Powerful contextual positioning’ relates to our earlier discussions about power in relational terms (Avelino and Wittmayer 2016), that is a PSO’s real influence over other external actors in the system and over the institutional framework of the system.

The second consideration has to do with public leadership. All organisations, including public sector organisations, have leaders, and the exercise of leadership is relevant to the functioning of the organisation. When dealing with transformative capacity in sustainability transitions and defining the roles of the PSO, public leadership matters. This has to do not only with the interpersonal skills of the individual leaders as they motivate others and set specific organisational goals; it has also to do with the overall organisational dimension of leadership where employees are (self-)aligned in working synergistically. This is a reason why public leadership (or the lack thereof) in sustainability transitions is an emerging topic (Lindvall 2023; Braams, Wesseling et al. 2022). It is worth noting at this point the distinction in the public administration literature between political leadership, that is exercised by (democratically elected) politicians and political groups, and public leadership, that is exercised by managers in public sector organisations. This paper focuses on the latter, and therefore we are more concerned with public leadership than with political leadership. We acknowledge that political leadership is important in sustainability transitions. When such political leadership results in clear and ambitious climate targets, this becomes a political mandate which is a fundamental resource for PSOs transformative capacity (see Section 5.2). In other cases, however, political leadership might not engage, leaving PSOs with a different room for manoeuvre which would require mobilizing implicit rather than explicit mandates, and unfolding specific directional roles circumventing political apathy or resistance. This has to do with the relationship and tensions between political and public authorities.

变革性变化的早期阶段，PSO可能充当发起者和领先用户，如上所述。然而，PSO并不总是主导者，因为其他路由社会或市场行为者引领。在这种情况下，PSO的变革能力角色主要是作为促进者（鼓励他人参与）并防止他人（或自身）成为障碍。一些PSO的核心地位在这方面至关重要。例如，在航运业中，可持续转型涉及私人船东将化石燃料船舶替换为使用零排放燃料的新船舶。海事监管机构是PSO，负责批准船舶以确保海事安全。如果没有它们的批准，任何零排放燃料船舶都无法启航，因此PSO对于绿色技术在航运业的部署至关重要。现有法规适应化石燃料技术，而海事监管机构的法律技术专长对于重新定义监管框架至关重要，它们在国际和国家监管层面充当早期谈判者。一旦框架明确并确立以支持新燃料，这些机构就开始对每艘新船进行审批工作。海事监管机构需要调动新的资源（例如关于新技术的新数据和外部专业知识），同时发展自身的内部分析能力以参与审批工作。在转型后期，当零排放船舶投入运营时，海事机构验证并监控安全。这要求该机构发展另一套操作能力，因为这些船舶的安全运营与化石燃料船舶截然不同。总之，海事监管机构在转型路径的不同阶段中的变革性角色、资源和能力各不相同（Taudal Poulsen和Borrás）。

大量文献强调，变革性转型很少是一个线性、无冲突或无成本的过程。根据紧张局势如何以及何时展开，公共部门组织可能扮演不同的角色，致力于最小化这些紧张局势，以避免转型被延误或偏离轨道。在早期阶段，公共部门组织可能扮演不同对手之间的调解者角色。它也可能作为特定方法的推动者，积极支持争议中的一方。无论哪种情况，无论是作为调解者还是特定一方的推动者，公共部门组织都需要调动其自身的合法性资源（基于其致力于公共利益的可信度），并构建令人信服的叙事。这有时可能并不容易。同样，它需要强大的沟通和协调能力来管理这种沟通。之后，公共部门组织还可能扮演缓解者的角色，寻求补偿转型负面影响的方法。这可能需要公共部门组织调动财务资源和运营能力来进行补偿，以及数据资源和分析能力，以便基于证据的干预措施确定哪些社会群体受影响最大以及如何最好地补偿他们。

上述所有内容表明，转型过程的阶段与公共部门组织的变革能力之间可能存在共同演化关系，两者都随时间而变化。尽管我们看到它可能随时间变化，但我们不能假设公共部门组织的变革能力会自动扩展和增长。这引出了本文关于构建变革能力的最后一个理论思考。

6.3. 建设公共部门组织的变革能力

第三个理论思考关乎变革能力的建设。我们以公共部门组织的角色为起点，假设在实施有目的变革代理时，这些角色是公共部门组织变革能力的必要条件。据此，我们可以假设公共部门组织变革能力的建设并非自动过程。在某种程度上，变革能力是在公共部门组织内部自然构建的，当然也与该组织运作的更广泛治理和公共行政环境相关。因此，这里可能涉及两个进一步的考虑：能力建设作为其构成要素逐步扩展的过程；以及能力建设作为与公共领导力相关的过程。

首先，我们可以理论化，变革能力是通过扩展公共部门组织（PSO）自身构成要素（即角色、资源和能力）的互动过程而建立的。'这些要素的互动意味着它们可能（也可能不）逐渐相互支持，在广度和强度上不断扩展。二元性作为利用能力和探索能力的概念，在这方面可能值得探讨（Gieske, van Buuren et al. 2016）。建立变革能力不仅是PSO'对外部社会政治需求的机械性响应，以执行集体目标。在某种程度上，这是PSO（由PSO自身或其上级部门–如部委等触发）的一种战略举措，旨在创造更强大的情境定位，以实施变革行动。“强大的情境定位”与我们之前关于关系性权力的讨论相关（Avelino and Wittmayer 2016），即PSO'对系统中其他外部行为者以及系统制度框架的实际影响力。

第二个考虑因素与公共领导力有关。所有组织，包括公共部门组织，都有领导者，领导力的行使与组织的运作相关。当涉及可持续转型中的变革能力并定义公共部门组织的角色时，公共领导力至关重要。这不仅关乎个体领导者激励他人和设定特定组织目标的人际交往能力；还关乎领导力的整体组织维度，即员工在协同工作中（自我）协调一致。这就是为什么公共领导力（或其缺失）在可持续转型中成为一个新兴话题（Lindvall 2023; Braams, Wesseling et al. 2022）。在此值得指出的是，公共行政文献中区分了由（民主选举的）政治家和政治团体行使的政治领导力，以及由公共部门组织管理者行使的公共领导力。本文关注后者，因此我们更关心公共领导力而非政治领导力。我们承认政治领导力在可持续转型中很重要。当这种政治领导力带来明确且雄心勃勃的气候目标时，这就成为一项政治授权，是公共部门组织变革能力的基本资源（参见第5.2节）。但在其他情况下，政治领导力可能不参与，这给公共部门组织留下了不同的运作空间，要求调动隐性而非显性的授权，并展开具体的导向性角色以规避政治冷漠或阻力。这与政治和公共权威之间的关系和紧张有关。

7. Conclusions

Public sector organisations (PSOs) are central to sustainability transitions, but the literature on transformative capacity of PSOs remains disjointed. To drive forward the emerging research agenda of PSO transformative capacity in sustainability transitions, we conducted a review of the literature, identified the various conceptual approaches and built a higher-order conceptual framework. We carried out a scope review of 121 selected publications to capture key conceptualisations in three literature strands, analysed their strengths and shortcomings and identified key insights in order to advance theorising and conceptual development.

Our findings show two key insights. First, we identified two main perspectives on capacity: one that defines capacity as the potential of an organisation based on its resources, defined by its situation in a governance system; and another that defines capacity in action-oriented terms as the organisation’s own abilities to enact transformative change. The latter perspective is associated with capability, a notion that denotes the specific routines of an organisation to perform diverse tasks. Acknowledging that both are crucial, we suggest that the capacity of an organisation is about the interaction between its resources (as situated assets internal and external to the organisation) and its abilities (as the routines, practices, and procedures of the organisation to use those resources). See Fig. 2.

The second key insight has to do with the *transformativeness* in transformative change. Considering that sustainability transitions require path-breaking change, the literature takes a process-oriented perspective on transformation. For this reason, the literature largely understands transformative agency as the purposeful and directional enactment of change. We elaborate on these key findings, suggesting to conceptualise transformative capacity as the interaction between a public sector organisation’s enactment of directional roles, its situated resources, and its abilities (see Fig. 3).

In terms of theoretical implications and contributions, this article elaborates on three theoretical considerations resulting from the above. First, that the transformative capacity of a PSO is neither static nor binary, but is defined by the interactions between roles, resources, and abilities, taking many possible different forms. How can we conceptualise that diversity of interactions? We take as the starting point that roles are necessary conditions, and resources and abilities are sufficient conditions for a PSO’s transformative capacity. A PSO’s enactment of change agency to overcome the heavy barriers towards sustainability transitions require directionality, and therefore the organisation’s own interpretation and purposeful enactment of various (new) roles that, combined, might be more or less transformative. These combinations of roles require the mobilisation of specific situated resources and organisational abilities. The article provides examples of these combinations of roles and their interactions with specific resources and abilities in Section 6.1.

The second theoretical consideration is the co-evolution between a PSO’s transformative capacity and sustainability transitions’ pathways. There is a widespread acknowledgement in the literature that sustainability transitions do not follow one single pathway, and that transitions are not free from costs and opposition. The transformative capacity of PSOs vary along this diversity of pathways and across time. In different transition pathways, a PSO will have different roles. Depending on how those transformative roles are combined, the PSO will require diverse sets of situated resources and organisational abilities. Likewise, the different phases of transitions’ pathways need various combinations of roles, resources, and abilities. Hence, we propose the theoretical consideration that there might be a co-evolution between the transformative capacity of the PSO, and the types of pathways and of its phases. However, we cannot assume that such co-evolution happens automatically.

This brings us to our third theoretical consideration about building transformative capacity. We suggest that a PSO’s transformative capacity-building is the process by which its constitutive elements are expanded; and we suggest that public leadership is important in that regard. Building transformative capacity in a PSO is not only a matter of providing it with more resources. It is also an inside-out

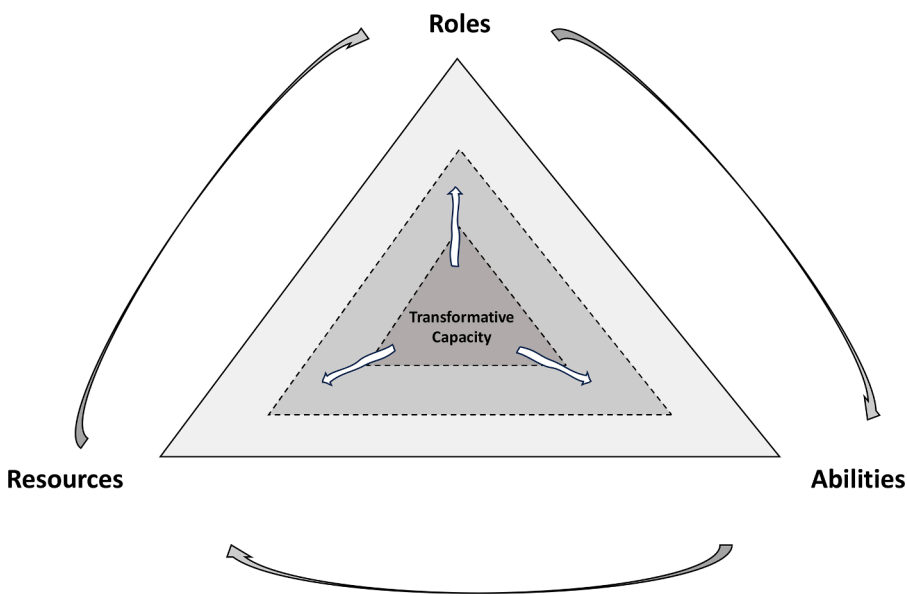


Fig. 3. The three constitutive elements of public sector transformative capacity.

7. 结论

公共部门组织在可持续转型中处于核心地位，但关于公共部门组织变革能力的文献仍然零散。为推进公共部门组织在可持续转型中变革能力的新兴研究议程，我们进行了文献综述，识别了各种概念方法，并构建了一个高阶概念框架。我们对121篇精选出版物进行了范围审查，以捕捉三个文献流派中的关键概念化，分析了它们的优势与不足，并识别了关键见解，以推进理论化和概念发展。

我们的发现揭示了两个关键见解。首先，我们识别出关于能力的两种主要视角：一种将能力定义为基于资源的组织潜力，由其治理系统中的位置所决定；另一种以行动为导向，将能力定义为组织自身实施变革的能力。后一种视角与“能力”概念相关联，该概念指代组织执行不同任务的具体惯例。认识到两者都至关重要，我们认为组织的能力是其资源（作为组织内外部的情境资产）与其能力（作为组织使用这些资源的惯例、实践和程序）之间的互动。参见图2。

第二个关键见解与变革中的变革性有关。考虑到可持续转型需要突破性变革，文献采用了过程导向的视角来看待转型。因此，文献在很大程度上将变革能动性理解为有目的、有方向的变革实施。我们详细阐述了这些关键发现，建议将变革能力概念化为公共部门组织的方向性角色实施、其情境资源及其能力之间的互动（参见图3）。

在理论意义和贡献方面，本文阐述了上述内容引发的三个理论思考。首先，公共部门组织的变革能力既非静态也非二元，而是由角色、资源和能力之间的相互作用所定义，并呈现多种可能的不同形式。我们如何概念化这种多样化的相互作用？我们以角色是必要条件、资源和能力是充分条件作为起点，来理解公共部门组织的变革能力。公共部门组织在克服可持续发展转型的重大障碍时，其变革能动性的发挥需要方向性，因此组织自身对各种（新）角色的解读和有意实施，可能更具或更不具变革性。这些角色的组合需要调动特定的情境资源和组织能力。本文在6.1节提供了这些角色组合及其与特定资源和能力相互作用的示例。

第二个理论思考是公共部门组织的变革能力与可持续发展转型路径之间的共同演化。文献中普遍承认，可持续发展转型并非遵循单一路径，且转型并非没有代价和反对。公共部门组织的变革能力随路径多样性和时间而变化。在不同的转型路径中，公共部门组织将扮演不同的角色。根据这些变革角色的组合方式，公共部门组织需要多样化的情境资源和组织能力。同样，转型路径的不同阶段也需要不同的角色、资源和能力组合。因此，我们提出理论思考：公共部门组织的变革能力与路径类型及其阶段之间可能存在共同演化。然而，我们不能假设这种共同演化会自动发生。

这引出了我们关于建设变革能力的第三个理论思考。我们认为，公共部门组织的变革能力建设是其构成要素扩展的过程；并且我们提出公共领导力在这方面至关重要。建设公共部门组织的变革能力不仅是为其提供更多资源的问题，也是其由内而外的

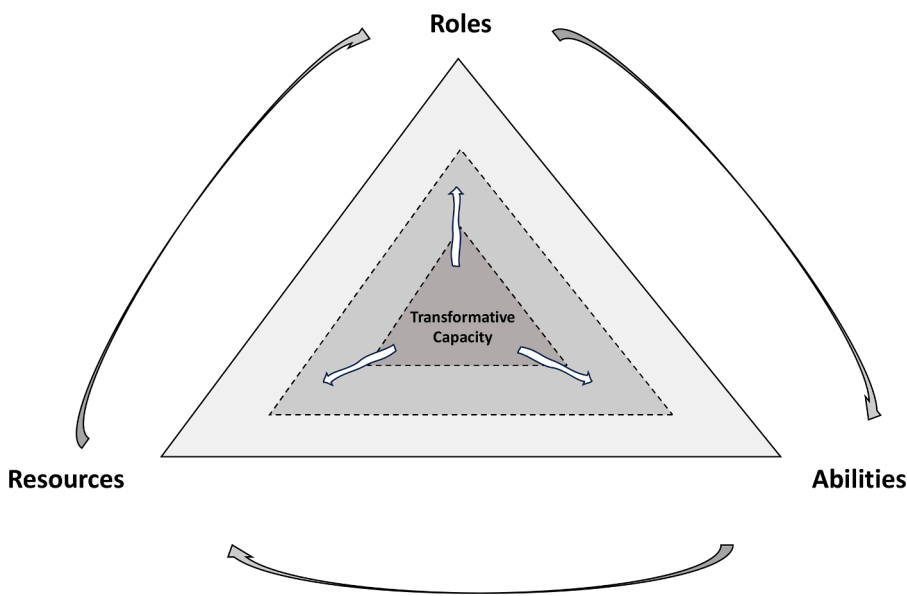


图3. 公共部门转型能力的三个构成要素。

move of the PSO itself, seeking to define its own roles, resources, and abilities, to gain a stronger positioning in a specific sustainability transition pathway. This is why the leadership in the PSO is important. Public leadership is not only about the interpersonal skills of the leading individual motivating and setting courses of action, but also about the (self-)alignment of employees towards transformative change.

Overall, our higher-order conceptual framework of PSOs transformative capacity, based on roles, resources, and abilities, offers the opportunity for a more granular empirical research endeavour to understand what specific interactions of transformative capacity elements are at play. The three theoretical considerations elaborated in the previous section of this article offer concrete clues towards a future research agenda based on a holistic framework of transformative capacity. Such an agenda is an empirical research endeavour asking questions about how the roles, resources, and abilities of public sector organisations interact in the enactment of change agency; asking questions about the co-evolution between transformative pathways and the unfolding of these roles, resources, and abilities; and asking questions about how transformative capacity is built through time. Each of these three questions brings forward a number of aspects that deserve careful study. Comparative studies could bring key insights in this regard. For example, studying variations in the transformative capacity of similar public sector organisations (i.e. two municipal public organisations) in the same socio-political and economic context (in the same country, and with similar urban system features) and in the same socio-technical system (energy transition). Variations of their transformative capacity will indicate clues about specific conditions, contrasting expectations about what roles/resources/routines are essential for transformative capacity. A second line of research could focus on barriers and lock-ins, asking how PSOs use transformative capacity to overcome different types of barriers and lock-ins in current socio-technical systems? Such a study could take the starting point in the nature of the barriers, and examine comparatively PSOs’ capacity as change agents, or their limits thereof. A third avenue for a future research could be to compare transformative capacity across socio-technical systems. Key questions are how and to what extent the nature of these socio-technical systems matters for the exercise and development of transformative capacity as change agency. A fourth research agenda could empirically examine the impact of various leadership strategies to enhance transformative capacity, given the relevance of public leaders and managers to define courses of organisational action. Last, but not least, a fifth research agenda could investigate the transformative capacity of public sector organisations that are facing particularly severe challenges of resource scarcity. This is particularly relevant for public sector organisations that have experienced a rapid resource reduction the past decades, and/or that have always been endemically resource-constrained due to wider socio-economic challenges (particularly low and middle-low income countries).

At the practical level, this conceptual framework could be adapted to a self-reflective tool for public leaders and civil servants, who are searching for concrete clues to make their organisations fit-for-purpose in specific sustainability transitions’ pathways. More concretely, they could use it to reflect upon and identify specific elements (roles, resources, abilities) that need to be strengthened and changed, allowing for a more targeted and context-embedded capacity-building efforts in public sector organisations.

CRediT authorship contribution statement

Susana Borrás: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stine Haakonsson:** Writing – review & editing, Visualization, Formal analysis. **Christian Hendriksen:** Writing – review & editing, Validation, Formal analysis. **Francesco Gerli:** Writing – review & editing, Validation, Formal analysis. **René Taudal Poulsen:** Writing – review & editing, Validation, Formal analysis. **Trine Pallesen:** Formal analysis. **Lucas Somavilla Croxatto:** Formal analysis. **Susanna Kugelberg:** Validation. **Henrik Larsen:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data is provided in the appendices

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Last but not least, we gratefully acknowledge as well the help of Fiona Frankel and Clara Loodh, our two dedicated student assistants.

自我推动，寻求定义自身角色、资源和能力，以在特定的可持续发展转型路径中获得更强定位的过程。这就是为什么公共部门组织的领导力很重要。公共领导力不仅涉及领导者个人的人际交往能力、激励和制定行动路线，还涉及员工对变革性变革的（自我）协同。

总体而言，我们基于角色、资源和能力的公共部门组织变革能力高阶概念框架，为更细粒度的实证研究提供了机会，以探究变革能力要素的具体相互作用。本文前一节阐述的三个理论思考，为基于变革能力整体框架的未来研究议程提供了具体线索。这一议程是一项实证研究任务，探讨公共部门的角色、资源和能力如何在变革推动中相互作用；探讨变革路径与这些角色、资源和能力的展开之间的共同演化；以及探讨变革能力如何随时间构建。这三个问题各自提出了一系列值得深入研究的方向。比较研究可以带来关键见解。例如，在同一社会政治和经济背景（同一国家，具有类似城市系统特征）以及同一社会技术系统（能源转型）中，研究类似公共部门组织（如两个市级公共组织）变革能力的差异。其变革能力的差异将揭示具体条件的线索，对比关于哪些角色/资源/常规对变革能力至关重要的预期。第二条研究路线可以关注障碍和锁定效应，探究公共部门组织如何利用变革能力克服当前社会技术系统中的各类障碍和锁定效应？此类研究可以以障碍的性质为起点，比较研究公共部门组织作为变革推动者的能力或其局限性。第三条未来研究路径可以是跨社会技术系统比较变革能力。关键问题是这些社会技术系统的性质在多大程度上影响变革能力作为变革推动力的行使和发展。第四条研究议程可以实证检验不同领导策略对增强变革能力的影响，鉴于公共领导者和管理者在确定组织行动方向方面的重要性。最后但同样重要的是，第五条研究议程可以调查资源匮乏挑战特别严峻的公共部门组织的变革能力。这对于过去几十年经历快速资源削减，或由于更广泛的社会经济挑战（特别是低收入和中低收入国家）而长期资源受限的公共部门组织尤为相关。

在实践层面，这一概念框架可改编为公共领导者和公务员的自我反思工具，他们正在寻找具体线索，以使自己的组织在特定的可持续转型路径中达成预期目标。更具体地说，他们可以利用它来反思并确定需要加强和改变的具体要素（角色、资源、能力），从而在公共部门组织中开展更具针对性和情境化的能力建设努力。

CRediT作者署名声明

Susana Borr´as: 写作 – 审阅与编辑、写作 – 初稿、可视化、验证、方法论、经费获取、正式分析、数据管理、概念化。**Stine Haakonsson:** 写作 – 审阅与编辑、可视化、正式分析。**Christian Hendriksen:** 写作 – 审阅与编辑、验证、正式分析。**Francesco Gerli:** 写作 – 审阅与编辑、验证、正式分析。**Ren´e Taudal Poulsen:** 写作 – 审阅与编辑、验证、正式分析。**Trine Pallesen:** 正式分析。**Lucas Somavilla Croxatto:** 正式分析。**Susanna Kugelberg:** 验证。**Henrik Larsen:** 验证。

利益冲突声明

作者声明，他们不存在可能影响本文所述工作的已知竞争性财务利益或个人关系。

数据可用性

所有数据均在附录中提供

致谢

我们感谢丹麦公共机构“丹麦独立研究基金”对CAPACITOR项目的慷慨资助，资助编号为0217–00076B

我们同样衷心感谢哥本哈根商学院的两位专业图书馆员在搜索脚本方面给予的帮助：Liv Bjerger Laursen 和 Mette Bechmann。

最后但同样重要的是，我们衷心感谢菲奥娜·弗兰克尔和克拉拉·洛德两位尽职的学生助理的帮助。

Appendix 1: Detailed methodology

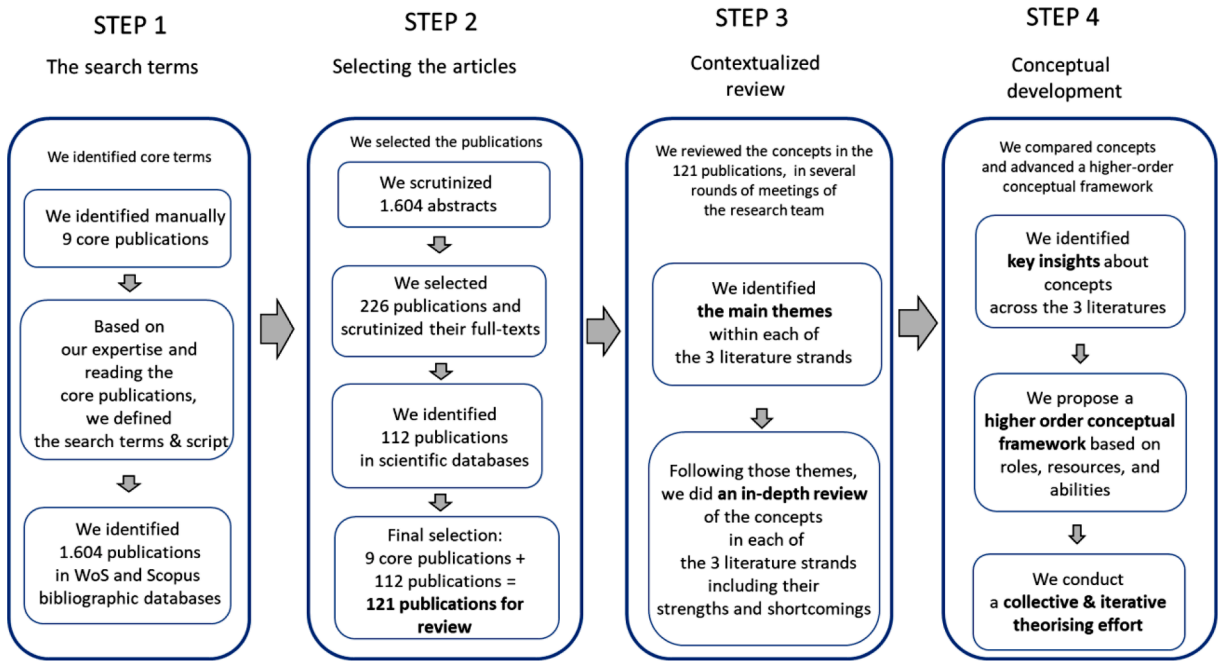
General about rigor, internal consistency, and relevance

As mentioned in [Section 2](#), we have paid careful attention to ensure the methodological quality of this theoretical literature review, and in this regard, we have followed the specific approach suggested by experts in literature reviews. More concretely, we have secured the rigor, the internal consistency and the relevance of our literature review in all its process.

Rigor refers to the thoroughness of the documentation about the search, selection, and synthesis process of the literature review. This refers to the soundness in the choices made along the way (Paré, Trudel et al. 2015). To secure the rigor of this review, we followed specific steps, and made careful decisions about screening, selection, organization, and reporting of the reviewed articles. Below we provide a detailed account of the different steps and the criteria used in each of them, showing that the whole process has followed carefully designed criteria in a rigorous manner. **Internal consistency** is another important aspect of the methodological quality of literature reviews. It refers to the consistency of the processes undertaken in relation to the features that characterize the type of review chosen from the beginning. In this article, we follow the guidelines and features of a “theoretical review”, according to the approach suggested by Paré et al. (see [Section 2](#) of the paper). More concretely, these authors suggest that a theoretical literature review has the following features: it defines research questions with a broad scope, it undertakes a comprehensive search strategy, it uses conceptual and empirical primary sources, it defines an explicit study selection, and it uses content analysis or interpretative methods for analyzing the findings. Our study is consistent with these features, as we have carefully followed each of them: our two research questions are broad and suitable for the overall aim of the review; we undertook a comprehensive search strategy (see below about Step 1); it used primary sources (scientific publications on the topic); defined explicit inclusion/exclusion criteria to guide the selection process (see below about Step 2); and underwent two cycles of coding, based structural coding (see below about Step 3). The third aspect that defines the methodological quality of the review has to do with the **relevance** of the literature review. This is related to the ability of the literature review article to answer convincingly its own research questions. We have defined these questions upfront (see [Section 1](#)). Our literature review is relevant, as we have carried out an encompassing analysis of the findings, which has allowed us to identify trends, as well as thinness in the literature. Following from that, we have developed a conceptual framework about the transformative capacity of public sector organizations based on three core elements: their roles, resources and skills; which serves to advance research in this field.

The Steps

As indicated in [Section 2](#) of the paper, and illustrated in [Fig. 1](#), our literature review followed 4 overall Steps:



In this appendix we provide additional information about the choices and criteria used in Steps 1 and 2, because these are the steps about the selection of the publications that were reviewed.

Regarding Steps 3 and 4 of the review process, [Sections 2, 3, 4, 5 and 6](#) of the paper provide all the details.

附录1: 详细方法

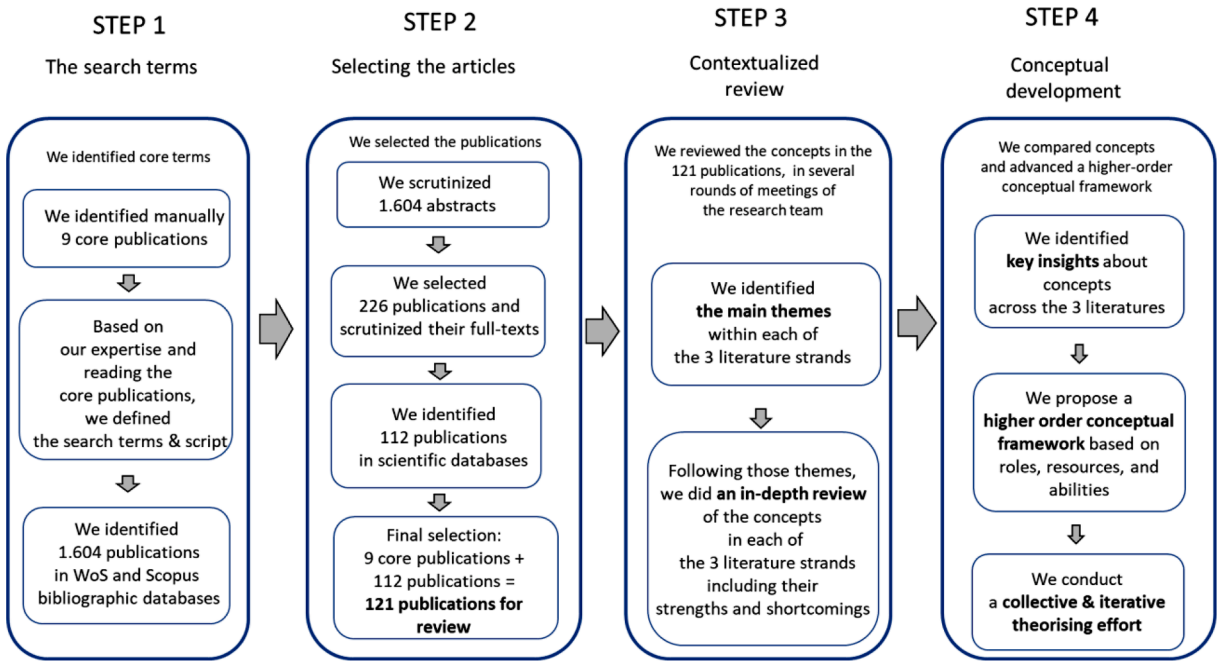
关于严谨性、内部一致性和相关性的概述

如[第2节](#)所述，我们已仔细确保这篇理论文献综述的方法论质量，并在这方面遵循了文献综述专家建议的具体方法。更具体地说，我们在整个过程中保证了文献综述的严谨性、内部一致性和相关性。

严谨性指的是文献综述的搜索、选择和综合过程的文档的全面性。这涉及沿途所做选择的合理性 (Paré, Trudel et al. 2015)。为确保本综述的严谨性，我们遵循了具体步骤，并对筛选、选择、组织和报告所评文献做出了审慎决策。下文将详细说明不同步骤及其使用的标准，表明整个过程遵循了精心设计的标准，以严谨的方式进行。**内部一致性**是文献综述方法质量的另一个重要方面。它指的是所进行的过程与最初选择的综述类型特征之间的一致性。在本文中，我们遵循了Paré等人建议的方法 (见[第2节](#))，采用“理论综述”的指导方针和特征。具体而言，这些作者认为理论文献综述具有以下特点：定义广泛范围的研究问题、采用全面的搜索策略、使用概念性和实证性的原始资料、明确研究选择标准，以及使用内容分析或解释性方法分析结果。我们的研究符合这些特征，因为我们认真遵循了每一条：我们的两个研究问题广泛且适合综述的总体目标；我们采用了全面的搜索策略 (见下文步骤1)；使用了原始资料 (该主题的科学出版物)；明确了纳入/排除标准以指导选择过程 (见下文步骤2)；并基于结构编码进行了两轮编码 (见下文步骤3)。影响综述方法质量的第三个方面是文献综述的**相关性**。这涉及文献综述文章令人信服地回答自身研究问题的能力。我们已提前定义了这些问题 (见[第1节](#))。我们的文献综述具有相关性，因为我们对研究发现进行了全面分析，从而识别了趋势和文献中的薄弱之处。在此基础上，我们基于三个核心要素：角色、资源和技能，提出了关于公共部门组织变革能力的概念框架，这有助于推动该领域的研究。

步骤

如指 在本文的[第2节](#)中进行了阐述，并在[图1](#)中进行了说明，我们的文献综述遵循了4个总体步骤 s:



在本附录中，我们提供了关于步骤1和2中所用选择和标准的额外信息，因为这些步骤涉及所审查出版物的筛选。关于审查过程的步骤3和4，[论文的第2节、3节、4节、5节和6节](#)提供了所有细节。

STEP 1: The search terms

In several iterative meetings, the research team and the expert librarians brainstormed and discussed the core search terms and their overall logical interactions. This served to provide a final list of search terms, and to undertake a script-based search in Scopus and Web of Science bibliographic databases with a time span from January 2000 to July 2023, yielding 1604 publications.

This time period is highly relevant due to the fact that the turn of the millennium represented a watershed in terms of how social, political, and academic debates have addressed the grand challenges of climate change and environmental sustainability.

The research team discussed the theme of the literature review from the perspective of their different backgrounds, their expertise in the field, and from the inspiration of a number of relevant background texts (see the list in Appendix 2).

In several iterative meetings, the research team generated a list of terms in order to guide the bibliographic search: the list contained terms that are synonyms to the three core topics of the search, namely, green/environmental sustainability, public sector/organizations, and capacity. The logic of the search was to find articles that are explicitly referring to all these three topics at once.

Interacting with two specialized librarians, the list of terms was further discussed and simplified. The terms were truncated with * to capture different word endings. The librarians recommended to conduct the search focusing on identifying those terms in the titles, abstracts, and keywords of articles, because this will yield outputs with lower levels of noise.^j

We followed as well the librarians’ recommendation to limit the search according to specific subject areas in the bibliographic databases, rather than to limit the search to specific journals. The reason is that this will allow to find relevant articles in a wider range of journals, which is relevant given the various strands of literature dealing with the topic. The subject areas in the bibliographical dataset Scopus were: environmental sciences, social science, business management and accounting, decision science, and multidisciplinary. We used similar subject areas in the other bibliographical dataset we used, namely, Web of Science.

Together with the librarians, we tested different versions of search scripts in order to identify which script yielded less noise. The script used:

((TITLE-ABS-KEY ((transition* OR transformation OR innovation*) W/2 (green OR sustainable)) OR TITLE-ABS-KEY (decarboni*) OR TITLE-ABS-KEY ("Socio-technic*" OR sociotechnic*) W/2 (system* OR niche*)) OR TITLE-ABS-KEY ("living lab*" OR livinglab) W/3 (sustain* OR green)) OR TITLE-ABS-KEY ((green OR sustainab* OR renew*) W/1 energy) OR TITLE-ABS-KEY (environment* W/2 (governance OR policy OR sustain*)))) AND (TITLE-ABS-KEY (((transform* OR change OR policy OR governance OR urban OR administrati* OR enforcement OR implement* OR transfer* OR regulat*) W/2 (capacit*)) OR "transformative capabili*" OR "dynamic capabili*")) AND (LIMIT-TO (SUBJAREA, "ENVI") OR LIMIT-TO (SUBJAREA, "SOCI") OR LIMIT-TO (SUBJAREA, "BUSI") OR LIMIT-TO (SUBJAREA, "DECI") OR LIMIT-TO (SUBJAREA, "MULT"))

We first run the script in the bibliographical database Scopus in the time period (2000-October 2021), and we re-run the same script in the database World of Science. After removing duplicates, and publications without clear author names or incomplete references, the final output for screening was 1.174 publications. In the Summer 2023 we re-run the same script to update the search covering the period from October 2021 to July 2023.

STEP 2: Selecting the publications

The inclusion and exclusion criteria were carefully discussed and refined to ensure the research team had a calibrated understanding of them. In the second sub-step we uploaded full-texts of the remaining 226 articles to a dedicated software program and engaged in a further selection based on reading the full papers. The full-text selection was based on the same exclusion criteria as in the screening (see Appendix 1), which resulted in a final selection of 112 studies for review (see the list of selected publications in Appendix 2).

Overall aim guiding the selection of the publications: We want to review the literature about the capacity of public sector organizations in sustainability green transitions, with the aim at developing a consistent review, and at advancing the conceptual framework in the field.

Two sub-steps:

- **Scrutinizing titles and abstracts:** We first screened/scrutinized the titles and abstracts of the 1.604 publications. Out of those, we selected 226 publications.
- **Scrutinizing and selecting full texts:** Thereafter, we screened/scrutinized those 226 publications, reading their full text. Out of those, we finally selected 112 final publications for the review.

Inclusion and exclusion criteria
We used the following inclusion and exclusion criteria for both sub-steps above.

a) Type of publication:

Inclusion:

^j Noise refers to search outputs that have a disproportionate amount of non-focused results.

步骤1：搜索术语

在多次反复会议中，研究团队和专家图书管理员共同集思广益，讨论了核心搜索词及其整体逻辑关系。这得出了最终的搜索词列表，并在Scopus和Web of Science书目数据库中进行了基于脚本的搜索，时间跨度从2000年1月至2023年7月，共检索到1604篇出版物。

这一时期具有高度相关性，因为千禧年之交是社会、政治和学术讨论如何应对气候变化与环境可持续性重大挑战的分水岭。

研究团队从各自不同的背景、该领域的专业知识以及一系列相关背景文本（见附录2列表）的启发出发，讨论了文献综述的主题。

在多次迭代会议中，研究团队生成了一份术语列表以指导书目搜索：该列表包含了三个核心搜索主题的同义词，即绿色/环境可持续性、公共部门/组织以及能力。搜索逻辑是找到同时明确提及这三个主题的文章。

在与两位专业图书馆员交流后，术语列表得到进一步讨论和简化。术语使用*进行截断以捕获不同的词尾。图书馆员建议在文章标题、摘要和关键词中搜索这些术语，因为这将产生噪声较低的输出结果。^j

我们也遵循了图书管理员的建议，根据书目数据库中的特定学科领域来限制搜索范围，而不是将搜索限制在特定期刊上。原因是这样做可以在更广泛的期刊中找到相关文章，考虑到涉及该主题的不同文献流派，这一点很有意义。书目数据集Scopus中的学科领域包括：环境科学、社会科学、商业管理与会计、决策科学以及多学科。我们在使用的另一个书目数据集（即Web of Science）中也使用了类似的学科领域。

我们与图书管理员一起测试了不同版本的搜索脚本，以确定哪个脚本产生的噪音更少。使用的脚本如下：

((TITLE-ABS-KEY ((转型* 或 转变 或 创新*) W/2 (绿色 或 可持续)) 或 TITLE-ABS-KEY (脱碳*) 或 TITLE-ABS-KEY ("社会技术*" 或 社会技术*) W/2 (系统* 或 利基*)) 或 TITLE-ABS-KEY ("生活实验室*" 或 生活实验室) W/3 (可持续* 或 绿色)) 或 TITLE-ABS-KEY ((绿色 或 可持续* 或 可再生*) W/1 能源) 或 TITLE-ABS-KEY (环境* W/2 (治理 或 政策 或 可持续*)))) 且 (TITLE-ABS-KEY (((转型* 或 变革 或 政策 或 治理 或 城市 或 行政* 或 执行 或 实施* 或 转移* 或 监管*) W/2 (能力*)) 或 "转型能力*" 或 "动态能力*")) 且 (限定于 (学科领域, "ENVI") 或 限定于 (学科领域, "SOCI") 或 限定于 (学科领域, "BUSI") 或 限定于 (学科领域, "DECI") 或 限定于 (学科领域, "MULT"))

我们首先在文献数据库Scopus中运行脚本，时间范围为2000年至2021年10月，然后在World of Science数据库中重新运行相同的脚本。在删除重复项以及没有明确作者姓名或不完整参考文献的出版物后，最终用于筛选的输出为1174篇出版物。2023年夏季，我们重新运行相同的脚本，以更新覆盖2021年10月至2023年7月期间的搜索。

步骤2：筛选出版物

纳入和排除标准经过仔细讨论和修订，以确保研究团队对其有校准的理解。在第二个子步骤中，我们将剩余的226篇文章的全文上传到专用软件程序，并根据阅读全文进行进一步筛选。全文筛选基于与初筛相同的排除标准（见附录1），最终选定112项研究进行综述（所选出版物列表见附录2）。

选择出版物的总体目标：我们希望回顾关于公共部门组织在可持续绿色转型中能力的文献，旨在形成一致的综述，并推进该领域的概念框架。

两个子步骤：

- **筛选标题和摘要：**我们首先筛选了1,604篇出版物的标题和摘要。从中我们选出了226篇出版物。
- **筛选并选择全文：**随后，我们筛选了这226篇出版物，阅读它们的全文。最终，我们选出了112篇最终出版物用于综述。

纳入和排除标准
上述两个子步骤我们均使用以下纳入和排除标准。

a) 出版物类型：

包容性：

^j 噪声指的是搜索结果中非聚焦结果比例过高的输出。

- We only include acknowledged publications: articles in journals, or chapters in books of academic nature.
- We only include articles older than 2000.
- We include publications with clear authorship.

Exclusion: Hence we exclude opening letters in special issues (those too general, not really providing in-depth conceptual approach), editorial letters (too generic), grey publications (f.ex. assessment reports, think-tank reports, etc.), or full books (we are interested in the specific chapters conceptualizing capacity); we exclude non-authored publications (with no clear author, or no specific author names); we exclude publications earlier than 2000.

a) **Thematic assessment:**

We include theoretical, conceptual and empirical articles (we will review the way in which they use the concepts ‘capacity’, ‘capability’, ‘competences’)

We include articles about sustainable transitions/green transformation/sustainability/green innovation, and similar; not about economic growth or transformation of industrial sectors.

We include articles dealing with ‘public sector organizations’, which can be understood as: governments, municipalities, public administrators, agencies, public actors, or similar.

We include articles on ‘governance’, ‘policy’, ‘organizations’ capacity.

Hence **we exclude** studies about economic growth/industrial transformation: we are interested in green/ecological/sustainability transformation; We exclude articles talking about capacity, capacity-building or capability-building at the very end of their analysis, and/or only in relation to ‘policy advice/practical implications’: f.ex.: a publication with generic remarks like “there is a need of capacity-building” in the conclusion section is to be excluded. We are interested in studies that explicitly analyze empirically or conceptually capacities/capabilities as a central part of their study and analysis, etc., not articles mentioning ‘capacity’ as policy advice.

We exclude as well publications which are not truly examining capacity, but mostly oriented towards assessing success, effects, effectiveness of policies, etc.: It is not the determination of empirical ‘success’ that interest us: it is the concepts and conceptual framework about capacity that is the aim of our review and hence selection criteria: We are interested in publications (theoretical and/or empirical) that explicitly develop/use concepts related to the capacity/capabilities in processes of sustainability green transitions.

Appendix 2: Reviewed Publications

The list of manually selected core publications in Step 1

1. Borrás, S. and J. Edler (2020). "The roles of the state in the governance of socio-technical systems’ transformation." Research Policy 49(5).
2. Breznitz, D., D. Ornston and S. Samford (2018). "Mission critical: the ends, means, and design of innovation agencies." Industrial and Corporate Change 27(5): 883–896.
3. Geels, F. W. (2020). "Micro-foundations of the multi-level perspective on socio-technical transitions: Developing a multi-dimensional model of agency through crossovers between social constructivism, evolutionary economics and neoinstitutional theory." Technological Forecasting and Social Change 152.
4. Haddad, C. R., V. Nakić, A. Bergek and H. Hellsmark (2022). "Transformative innovation policy: A systematic review." Environmental Innovation and Societal Transitions 43: 14–40.
5. Kivimaa, P., W. Boon, S. Hyysalo and L. Klerkx (2019). "Towards a typology of intermediaries in sustainability transitions: A systematic review and a research agenda." Research Policy 48(4): 1062–1075.
6. Köhler, J., F. W. Geels, F. Kern, J. Markard, E. Onsongo, A. Wieczorek, F. Alkemade, F. Avelino, A. Bergek, F. Boons, L. Fünf-schilling, D. Hess, G. Holtz, S. Hyysalo, K. Jenkins, P. Kivimaa, M. Martiskainen, A. McMeekin, M. S. Mühlemeier, B. Nykvist, B. Pel, R. Raven, H. Rohracher, B. Sandén, J. Schot, B. Sovacool, B. Turnheim, D. Welch and P. Wells (2019). "An agenda for sustainability transitions research: State of the art and future directions." Environmental Innovation and Societal Transitions 31: 1–32.
7. Teece, D. J., G. Pisano and A. Shuen (1997). "Dynamic capabilities and strategic management." Strategic Management Journal 18 (7): 509–533.
8. Weber, K. M. and H. Rohracher (2012). "Legitimizing research, technology and innovation policies for transformative change: Combining insights from innovation systems and multi-level perspective in a comprehensive ‘failures’ framework." Research Policy 41(6): 1037–1047.
9. Wu, X., M. Ramesh and M. Howlett (2018). Policy Capacity: Conceptual Framework and Essential Components. Policy Capacity and Governance. Assessing Governmental Competences and Capabilities in Theory and Practice. X. Wu, M. Howlett and M. Ramesh. Cham, Springer International Publishing: 1–25.

The list of the 112 publications selected using search terms and script:

- 我们仅收录公认的出版物：期刊文章或学术性质的书籍章节。
- 我们仅收录2000年之后的文章。
- 我们收录具有明确作者身份的出版物。

排除标准：因此，我们排除特刊中的开场信（过于泛泛，未提供深入的概念性方法）、编辑信（过于通用）、灰色出版物（例如评估报告、智库报告等）或整本书籍（我们关注的是具体章节中概念化的能力）；我们排除无作者身份（无明确作者或无具体作者姓名）的出版物；我们排除2000年之前的出版物。

a) **主题评估：**

我们纳入理论性、概念性和实证性文章（我们将回顾它们如何使用“容量”、“能力”、“胜任力”等概念）

我们纳入关于可持续转型/绿色转型/可持续性/绿色创新等的文章，而非关于经济增长或工业部门转型的文章。

我们纳入涉及“公共部门组织”的文章，这些组织可以理解为：政府、市政机构、公共管理者、机构、公共行为者等。

我们纳入关于“治理”、“政策”、“组织能力”的文章。

因此**我们排除**关于经济增长/产业转型的研究：我们关注绿色/生态/可持续转型；我们排除那些在分析末尾才提及能力、能力建设或能力构建的文章，以及/或仅与“政策建议/实践启示”相关的文章：例如，结论部分有“需要能力建设”等泛泛之言的出版物将被排除。我们关注的是那些在研究和分析中明确将能力/能力作为核心部分进行实证或概念性分析的研究，而不是那些将“能力”作为政策建议提及的文章。

我们还排除了并非真正研究能力，而主要致力于评估政策成功、效果、有效性等的出版物：我们感兴趣的不是实证上的“成功”，而是关于能力的概念和概念框架，这是本次综述及其选择标准的目标：我们关注那些明确开发/使用与可持续绿色转型过程中的能力/能力相关的概念（理论性和/或实证性）的出版物。

附录2：已审查的出版物

步骤1中手动选择的核心出版物列表

1. Borrás, S. 和 J. Edler (2020). "社会技术系统转型治理中的国家角色。" 《研究政策》49(5).
2. Breznitz, D., D. Ornston 和 S. Samford (2018). "关键任务：创新机构的宗旨、手段与设计。" 《工业与公司变革》27(5): 883–896.
3. Geels, F. W. (2020)。"社会技术转型多层次视角的微观基础：通过社会建构主义、演化经济学和新制度理论的交叉构建多维度能动性模型。" 《技术预测与社会变革》152.
4. Haddad, C. R., V. Nakić, A. Bergek 和 H. Hellsmark (2022). "变革性创新政策：系统综述。" 《环境创新与社会转型》43: 14–40.
5. Kivimaa, P., W. Boon, S. Hyysalo 和 L. Klerkx (2019)。"迈向可持续转型中介的类型学：系统综述与研究议程。" 《研究政策》48(4): 1062–1075.
6. Köhler, J., F. W. Geels, F. Kern, J. Markard, E. Onsongo, A. Wieczorek, F. Alkemade, F. Avelino, A. Bergek, F. Boons, L. Fünf-schilling, D. Hess, G. Holtz, S. Hyysalo, K. Jenkins, P. Kivimaa, M. Martiskainen, A. McMeekin, M. S. Mühlemeier, B. Nykvist, B. Pel, R. Raven, H. Rohracher, B. Sandén, J. Schot, B. Sovacool, B. Turnheim, D. Welch 和 P. Wells (2019)。"可持续转型研究议程：现状与未来方向。" 《环境创新与社会转型》31: 1–32.
7. Teece, D. J., G. Pisano 和 A. Shuen (1997)。"动态能力与战略管理。" 《战略管理杂志》18(7): 509–533.
8. Weber, K. M. 和 H. Rohracher (2012)。"为变革性变革的研发与创新政策正名：在综合'失败'框架中融合创新系统和多层次视角的见解。" 《研究政策》41(6): 1037–1047.
9. Wu, X., M. Ramesh 和 M. Howlett (2018)。政策能力：概念框架与基本要素。政策能力与治理：理论与实践中的政府能力与胜任力评估。X. Wu, M. Howlett 和 M. Ramesh 编。Cham, Springer International Publishing: 1–25。

使用搜索词和脚本筛选出的112篇出版物列表：

1. Aguilar, M.G., Jaramillo, J.F., Ddiba, D., Páez, D.C., Rueda, H., Andersson, K., Dickin, S., 2022. Governance challenges and opportunities for implementing resource recovery from organic waste streams in urban areas of Latin America: insights from Chía, Colombia. *Sustainable Production and Consumption* 30, 53–63. 10.1016/j.spc.2021.11.025

2. Anadon, L.D., Chan, G., Harley, A.G., Matus, K., Moon, S., Murthy, S.L., Clark, W.C., 2016. Making technological innovation work for sustainable development. *Proc Natl Acad Sci U S A* 113, 9682–9690. 10.1073/pnas.1525004113

3. Arshad, M., Yu, C.K., Qadir, A., Rafique, M., 2023. The influence of climate change, green innovation, and aspects of green dynamic capabilities as an approach to achieving sustainable development. *Environ Sci Pollut Res Int* 30, 71,340–71,359. 10.1007/s11356–023–27,343–0

4. Arts, B., Leroy, P., 2006. Institutional processes in environmental governance: Lots of dynamics, not much change?, in: Arts, B., Leroy, P. (Eds.), *Institutional Dynamics in Environmental Governance*. Springer, Dordrecht, The Netherlands, pp. 267–282. 10.1007/1–4020–5079–8_13

5. Avelino, F., Wittmayer, J.M., 2016. Shifting Power Relations in Sustainability Transitions: A Multi-actor Perspective. *Journal of Environmental Policy & Planning* 18, 628–649. 10.1080/1523908x.2015.1112259

6. Banjongprasert, A., 2022. Understanding the Adaptive Capacity of Actors in Environmental Governance: The Case of Trans-boundary Movement of Waste Electrical and Electronic Equipment. *Croatian International Relations Review* XXVIII, 62–89. 10.2478/CIRR-2022–0004

7. Barnes, M.L., Datta, A., Morris, S., Zethoven, I., 2022. Navigating climate crises in the Great Barrier Reef. *Global Environmental Change* 74. 10.1016/j.gloenvcha.2022.102494

8. Beck, S., Jasanoff, S., Stirling, A., Polzin, C., 2021. The governance of sociotechnical transformations to sustainability. *Current Opinion in Environmental Sustainability* 49, 143–152. 10.1016/j.cosust.2021.04.010

9. Benoit, P., Clark, A., Schwarz, M., Dibley, A., 2022. Decarbonization in state-owned power companies: Lessons from a comparative analysis. *Journal of Cleaner Production* 355. 10.1016/j.jclepro.2022.131796

10. Bettini, Y., Brown, R.R., de Haan, F.J., Farrelly, M., 2015. Understanding institutional capacity for urban water transitions. *Technological Forecasting and Social Change* 94, 65–79. 10.1016/j.techfore.2014.06.002

11. Bina, O., 2010. Environmental governance in China: Weakness and potential from an environmental policy integration perspective. *China Review* 10, 207–240.

12. Bissix, G., Rees, J.A., 2001. Can strategic ecosystem management succeed in multiagency environments? *Ecological Applications* 11, 570–583. 10.1890/1051–0761(2001)011[0570:CSEMSI]2.0.CO;2

13. Blythe, J.L., Cohen, P.J., Eriksson, H., Harohau, D., 2022. Do governance networks build collaborative capacity for sustainable development? Insights from Solomon Islands. *Environ Manage* 70, 229–240. 10.1007/s00267–022–01,644–5

14. Boer, H., 2013. Governing ecosystem Carbon. *Global Environmental Politics* 13, 123–143. 10.1162/GLEP_a_00201

15. Borgström, S., 2019. Balancing diversity and connectivity in multi-level governance settings for urban transformative capacity. *Ambio* 48, 463–477. 10.1007/s13280–018–01,142–1

16. Börzel, T.A., Buzogány, A., 2019. Compliance with EU environmental law. The iceberg is melting. *Environmental Politics* 28, 315–341. 10.1080/09,644,016.2019.1549772

17. Bos, J.J., Brown, R.R., 2014. Assessing organisational capacity for transition policy programs. *Technological Forecasting and Social Change* 86, 188–206. 10.1016/j.techfore.2013.09.012

18. Bowen, F., Panagiotopoulos, P., 2020. Regulatory roles and functions in information-based regulation: a systematic review. *International Review of Administrative Sciences* 86, 203–221. 10.1177/0,020,852,318,778,775

19. Bryan, T.K., 2016. Capacity for climate change planning: assessing metropolitan responses in the United States. *Journal of Environmental Planning and Management* 59, 573–586. 10.1080/09,640,568.2015.1030499

20. Castan Broto, V., Trencher, G., Iwaszuk, E., Westman, L., 2019. Transformative capacity and local action for urban sustainability. *Ambio* 48, 449–462. 10.1007/s13280–018–1086-z

21. Chaudhury, A., 2020. Role of intermediaries in shaping climate finance in developing countries-lessons from the green climate fund. *Sustainability (Switzerland)* 12. 10.3390/SU12145507

22. De Laurentis, C., Pearson, P.J.G., 2021. Policy-relevant insights for regional renewable energy deployment. *Energy, Sustainability and Society* 11. 10.1186/s13705–021–00,295–4

23. Dent, C.M., 2017. East Asia’s new developmentalism: state capacity, climate change and low-carbon development. *Third World Quarterly* 39, 1191–1210. 10.1080/01,436,597.2017.1388740

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28. Fischer, L.-B., Newig, J., 2016. Importance of Actors and Agency in Sustainability Transitions: A Systematic Exploration of the Literature. *Sustainability* 8. 10.3390/su8050476

1. Aguilar, M.G., Jaramillo, J.F., Ddiba, D., Páez, D.C., Rueda, H., Andersson, K., Dickin, S., 2022. 拉丁美洲城市地区有机废弃物资源回收的治理挑战与机遇：来自哥伦比亚Chía的见解。《可持续生产与消费》30, 53–63。10.1016/j.spc.2021.11.025

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